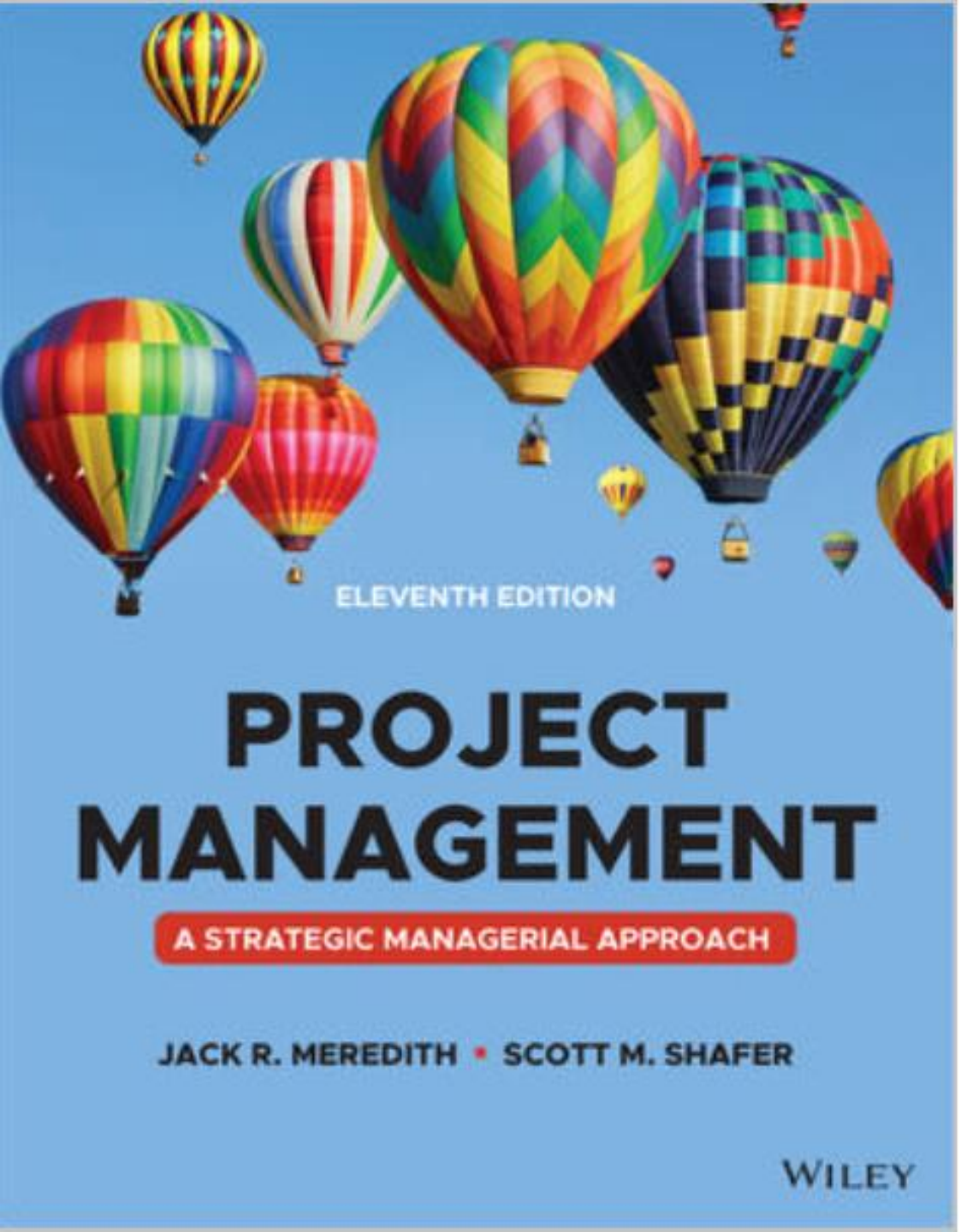

Class #8

Networks & Scheduling

Updated X15, Feb 2023 GK

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Next Topic

Learning Rate Recap

6:00pm (10min)

Mid-Term Exam Stats & Solutions

7:15

Entrepreneurship & Innovation

6:40pm (30min)

Network Diagrams & Gantt Charts

8:15

Network Solutions & Critical Path

Project Related & Examples

Learning Rate

Review (Class #5)

Actual time for the n-th unit, $L_n = S + T_n$

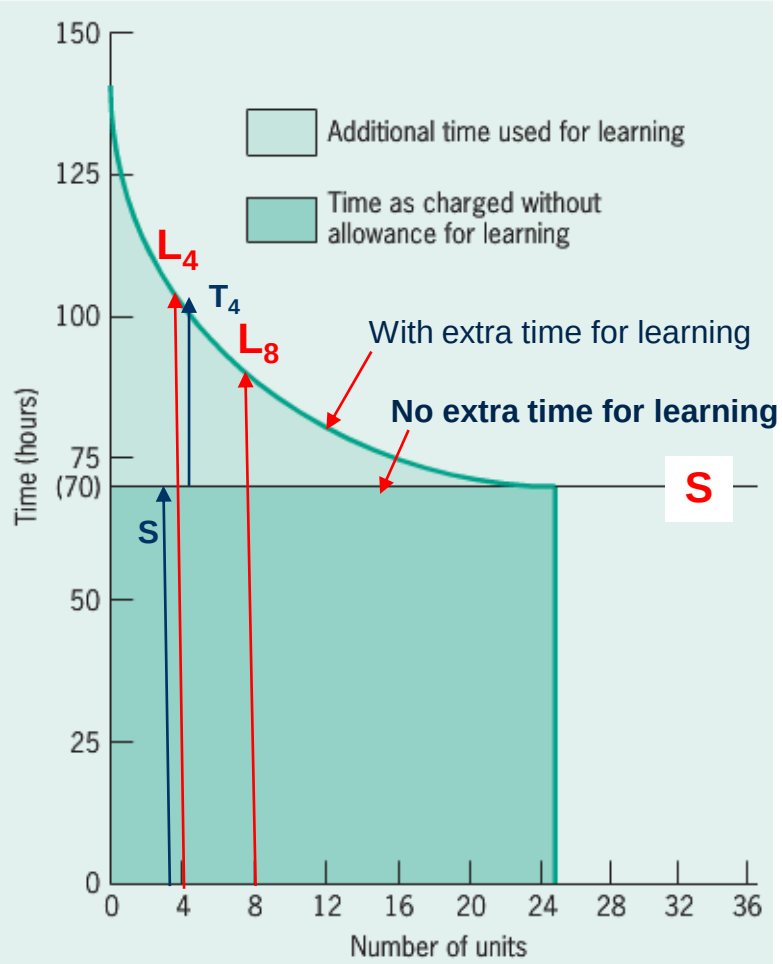
T_n = *Incremental* time to produce the *n*-th unit

R = learning rate

$$T_n = T_1 R^{\log_2 n} = T_1 n^{r \log_2 R}$$

T_1 = *Incremental* time to produce the first unit

$$L_n = S + T_1 R^{\log_2 n}$$



Review

Examples of Wright's Model

Assume R = 80% and $20/hr

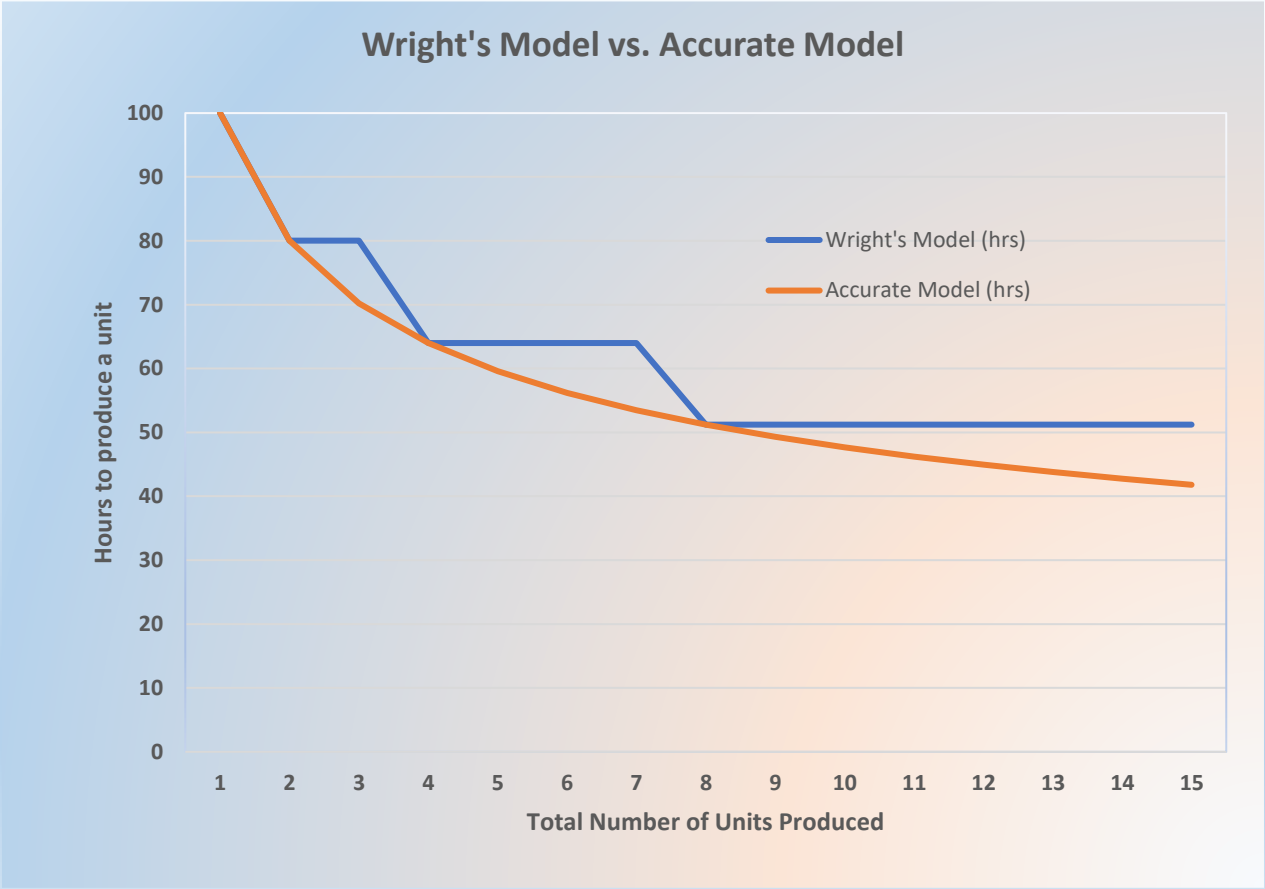
Unit produced (n)	TOTAL units produced	(Incremental) Labor hours *per unit* (T_n)	Incremental Labor hours	Total Labor hours	Cost per unit	Production Cost
1	1	100 (=T)	100 (for 1)	100 (for 1)	$2,000 (for 1)	$2,000
2	3	80 (=T*R)	160 (for 2 more)	260 (for 3)	$1,600 (for 2)	$3,200
4	7	64 (=T*R^2)	256 (for 4 more)	516 (for 7)	$1,280 (for 4)	$5,120
8	15	51.2 (=T*R^3)	409.6 (for 8 more)	925.6 (for 15)	$1,024 (for 8)	$8,192

- 80% learning curve r = log 0.8/log 2 = -0.09691/0.301 = -0.32196
- If first unit took 100 hrs, $L_n = 100 * n^{-0.32196}$
- If the "newbie" has to produce 7 units, the cost would be…
- But what is the *average cost per unit at the end?* (=925.6*20/15, $1,234/unit)

Review

Wright Model Approximation

Unit	Wright's Model (hrs)	Accurate Model (hrs)
1	**100**	**100.0**
2	**80**	**80.0**
3	80	70.2
4	**64**	**64.0**
5	64	59.6
6	64	56.2
7	64	53.4
8	**51.2**	**51.2**
9	51.2	49.3
10	51.2	47.6
11	51.2	46.2
12	51.2	44.9
13	51.2	43.8
14	51.2	42.8
15	51.2	41.8
16	102.4	102.4
	926	**851**



Agenda

Learning Rate Recap

6:00pm (10min)

Mid-Term Exam Stats & Solutions

6:10pm (30min)

7:10

Entrepreneurship & Innovation

Network Diagrams & Gantt Charts

Network Solutions & Critical Path

8:15

Project Related & Examples

Expectations previously set

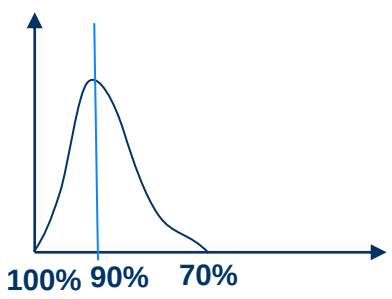
These rules have been applied universally for everyone. This uniform logic results to a *relative ranking* of grading.

1. **Multiple Choice [MC]:**
   - Give only one answer unless it is asked otherwise. Explanations might also be asked.
   - GRADING: Marking more than one answer, not answering directly, writing a story, etc → zero credit.
2. **Critical Thinking [CT]:**
   - Give a concise & clear answer -- do not lose time giving a long verbose answer; *you must answer the* question directly – indirect or non-understandable answers get no points
   - GRADING:
     - If you have answered exactly what matters the most → full credit
     - close enough → partial credit to my discretion.
     - not a direct answer, irrelevant answer, incorrect answer, incomprehensible answer → zero credit
3. **Numerical Problems [NP]:**
   - *You must answer the question directly in the WORD file*. Explain to the degree I can understand them. Correct answers with no explanations get zero points. Your Excel attachments must show your calculations. Regardless of your computations in the Excel spreadsheet, your final result must be included in the WORD file.
   - GRADING:
     a. Correct logic & correct result: → full credit
     b. Correct logic & incorrect results: → partial credit to my discretion.
     c. Partially correct logic (regardless of the result): → some credit to my discretion.
     d. Incorrect interpretation of problem, incorrect logic (regardless of the result); → some credit

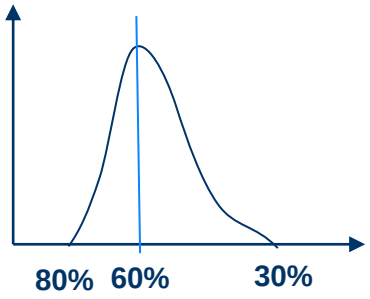
.

Clarification on Grading

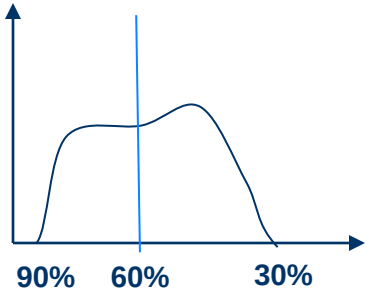
- *Everything is graded in an "absolute scale", that is,*
  - Each Homework question has specific answer(s) and it is either correct or partially correct, or not correct/answered.
  - Same is applicable for the Mid-Term and later-on for the Final Exam
  - The Project Report & Presentation will also follow a Rubric per section.
- *Each of the above is be normalized numerically at the end*
  - Class Participation is assessed subjectively at this point
  - Might have one or two outliers (low or high) which will be treated differently
  - Letter grades are determined based on University Guidelines



Class Stats on HW



Class Stats on MidTerm

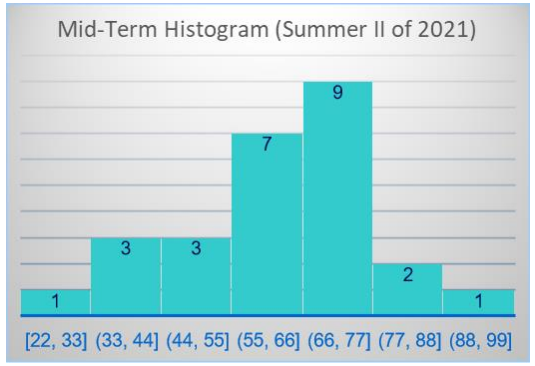


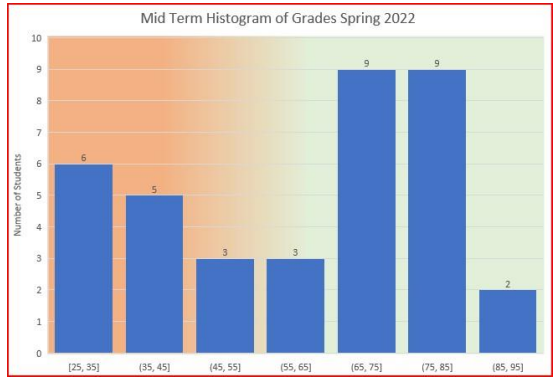
Class Stats on Final

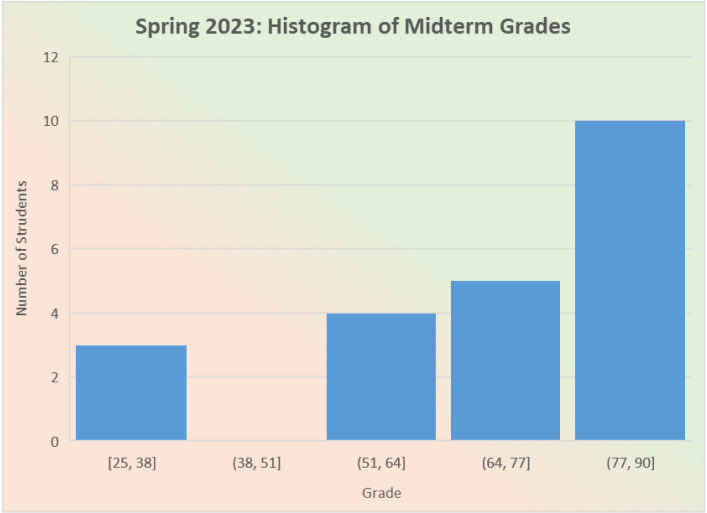
Normalization
at the end

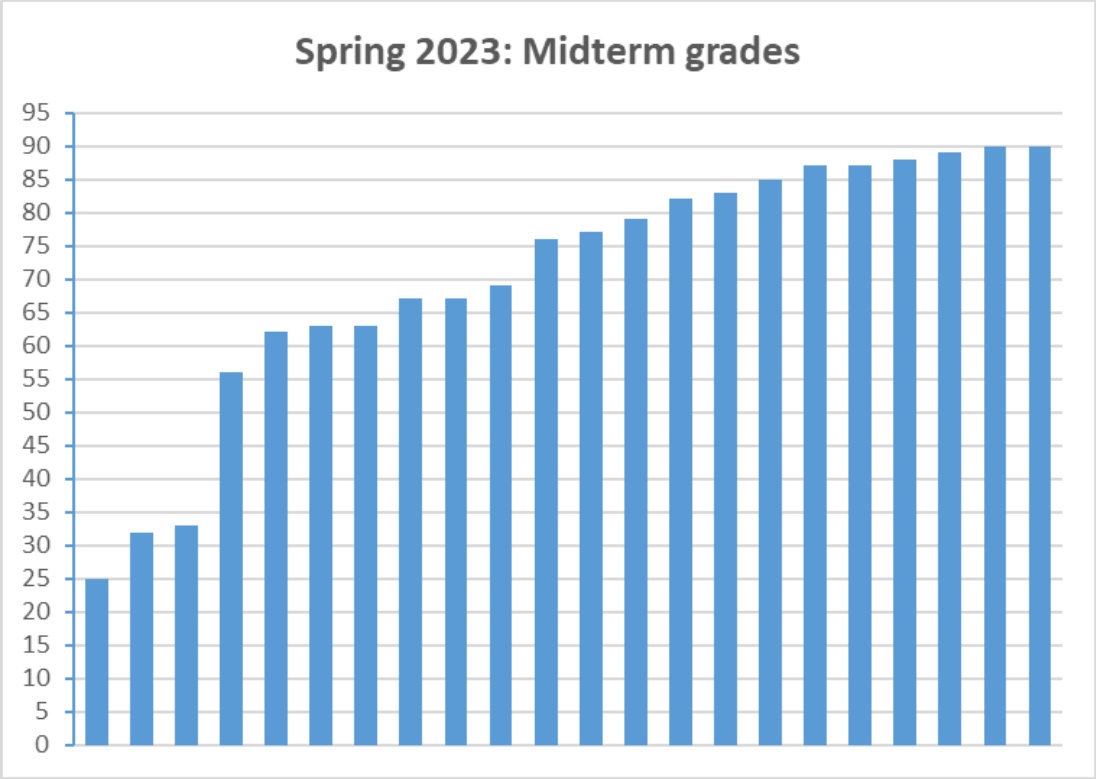
Relative ranking is preserved!

Mid-Exam Statistics









Summary of challenges

Question	Points	Average pts	%
[CT] ORG	8	6.8	85%
[MC] Tools	10	8.3	83%
[NP] Battery	12	9.6	80%
[MC] Maturity	6	4.8	80%
[CT] Projectized	9	6.6	74%
[CT] Neg	10	7.3	73%
[CT] SD	7	4.8	69%
[MC] APM	6	4.1	68%
[NP] Learning	12	7.4	62%
[NP] NPV	12	7.1	59%
[MC] Late	8	3.6	45%

Excellent news:

- 80% of the questions got >66% of points

Challenges:

1. You did not read the problem carefully.
2. You did not provide a clear answer in the WORD file.
3. You waisted time in lengthy “stories” and you run out of time

[CT] SD, 7pts

State Diagrams involve “states”. A “state” is: _____

(*Fill-in the definition of a “state”*)

States change on _____ which are always followed by _____.

(*Fill-in what happens as states transition to another state*)

[MC] APM, 6pts

Consider Traditional Project Management (i.e. Defined) versus APM (i.e. empirical).
Mark *all* answers below *not* relating to APM.

a. your team members are innovative
b. the Product Manager has clear requirements
c. you have a hard deadline
d. the WBS tasks can be identified
e. you must meet FDA approvals
f. your scope needs to be completed above 80%
g. you have unknown-known risks

Correct answers count points up. Incorrect answers count points down

[NP] Learning, 12pts

John spent 60min assembling the 5th unit incrementally to the time an expert typically spends. Next day he spent only 50min to assemble the 6th unit incrementally to the time an expert spends.

What is his learning rate?

You must show your calculations either inline or by attaching a spreadsheet. Regardless, you must include your final answer in this WORD file.

[NP] NPV, 12pts

The past two years, 2021 and 2022, the company made an investment of 20M and 40M respectively.
The sales forecast for the next 6 years (2023 etc) are 10M, 15M, 20M, 25M, 20M and 20M respectively.
The manufacturing and SG&A costs are about 20% of the annual revenues.
The current inflation rate is 5%

a. **[6pts] What is the time to break-even?**
b. **[6pts] What is the Performance Index at the 3rd year after starting the sales?**

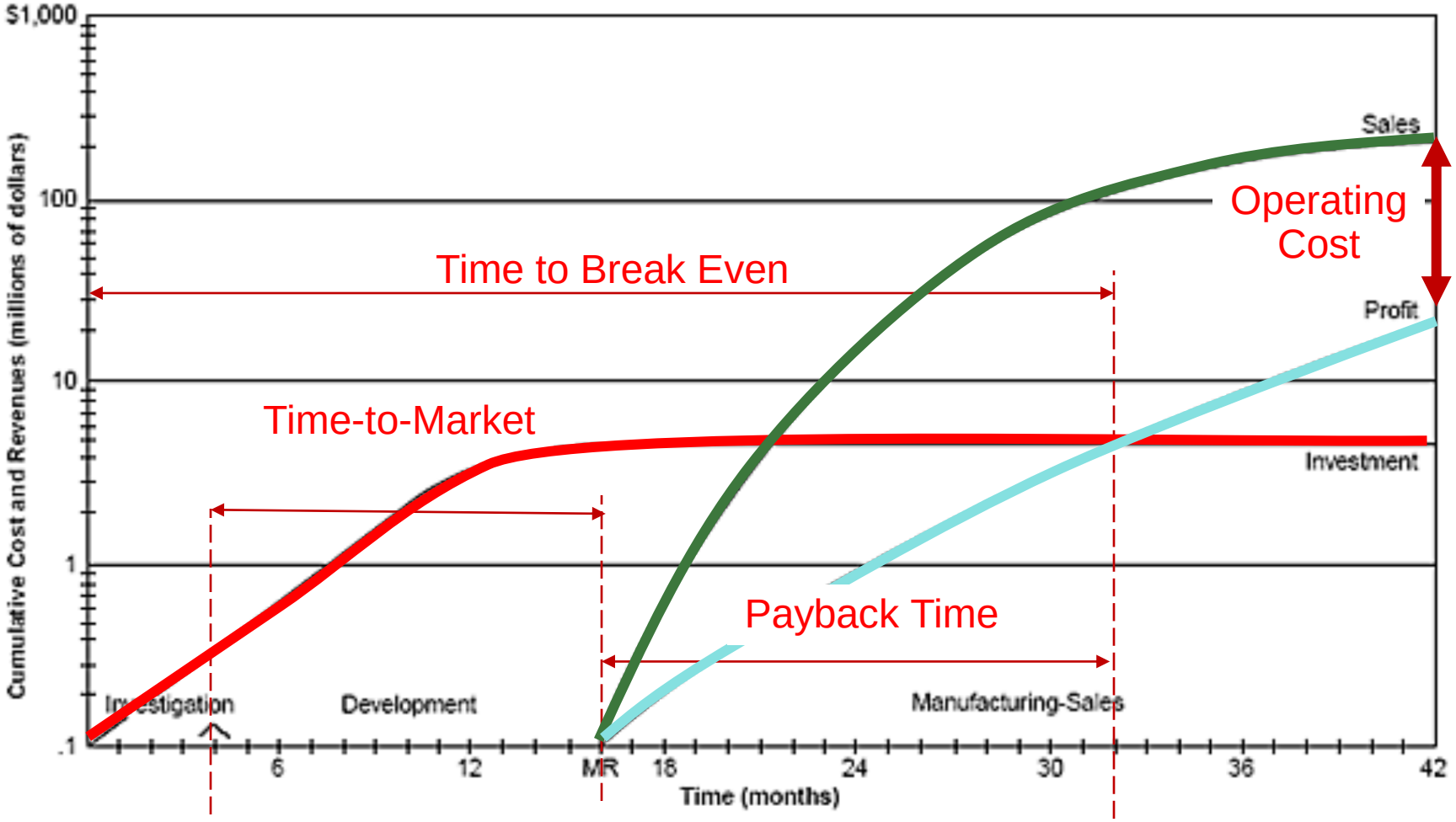
- *You must show your calculations by attaching a spreadsheet.*
- *Format your numbers to 2 or 3 significant digits – you will lose 2-points if you do not do this.*
- *You must include your direct and final answers in the WORD file. No points if answers are not included in WORD, regardless is shown in the Excel spreadsheet*
- ***Read and understand the question before answering***

NPV Solution

EoY	Year of Sales	Investment	OPS Cost	Revenue	NET	5% Scale of PV	PV	NPV gains	NPV	Notes
2021		-20			-20	1	-20			
2022		-40			-40	1	-40		-60.0	
2023	1		-2	10	8	1.05	7.6	7.6	-52.4	
2024	2		-3	15	12	1.10	10.9	18.5	-41.5	
2025	3		-4	20	16	1.16	13.8	32.3	-27.7	<-- 3rd year
2026	4		-5	25	20	1.22	16.5	48.8	-11.2	
2027	5		-4	20	16	1.28	12.5	61.3	1.3	<-- break even
2028	6		-4	20	16	1.34	11.9	73.3	13.3	

Shown 4-times, including the Review Deck

Project Cash Flow



https://hbr.org/1991/01/the-return-map-tracking-product-teams

[MC] Late, 8pts

Projects typically get completed late and/or go over-budget.

What is the *main* reason?

a. Too much scope creep
b. Incorrect NPV calculation
c. Insufficient upfront planning
d. An improperly constructed WBS
e. An unplanned risk event

Next Topic

Learning Rate Recap

Mid-Term Exam Stats & Solutions

7:15

Entrepreneurship & Innovation

6:30pm (30min)

Network Diagrams & Gantt Charts

7:20pm (30min)

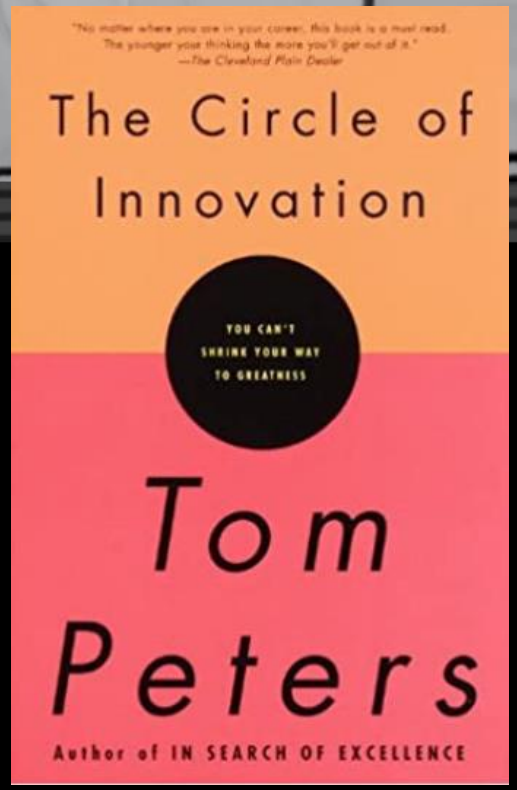
Network Solutions & Critical Path

8:15

Project Related & Examples

Startups

From HW#7



Lessons learned

Discussion about Entrepreneurship

- **Volunteers?**

Based on Homework #7, what is your take-away from Nancy's interview?

- Three bullets!
  - I learned …
  - I learned …
  - I learned …

Next Topic

Learning Rate Recap

Mid-Term Exam Stats & Solutions

6:10pm (20min)

7:15

Entrepreneurship & Innovation

Network Diagrams & Gantt Charts

7:20pm (30min)

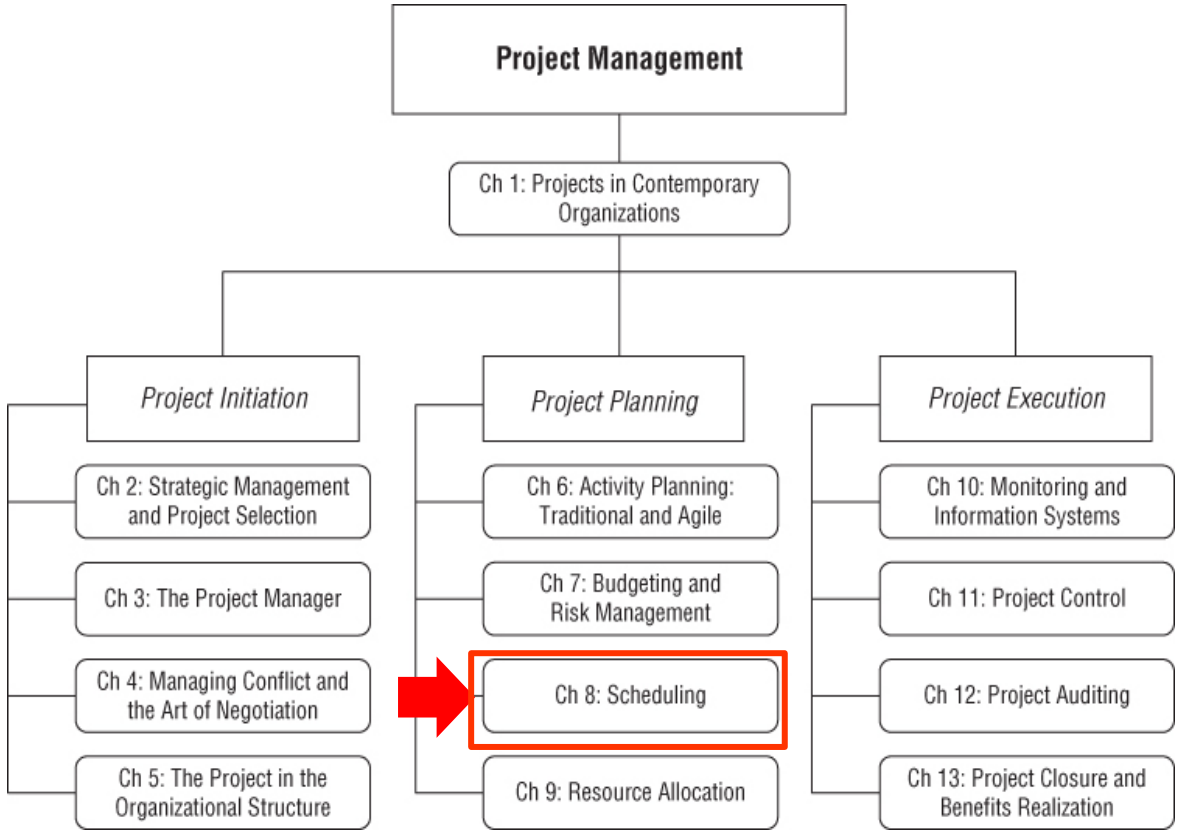
Network Solutions & Critical Path

7:50pm (25min)

8:15

Project Related & Examples

Today's Topic: Scheduling & Networks



Review

Portfolio→Programs→Projects: the Flow

- Strategic fit,
- Market opportunity
- Competitive position
- Financial benefit

Bz Plan

Charter

- Purpose
- Objectives (goals and priorities)
- Overview (HL managerial and technical approach)
- Schedules (key milestones)
- Resources (budget, contractors, equipment)
- Stakeholders
- Risk Management Plans
- Evaluation Methods

Project Plan

Tools

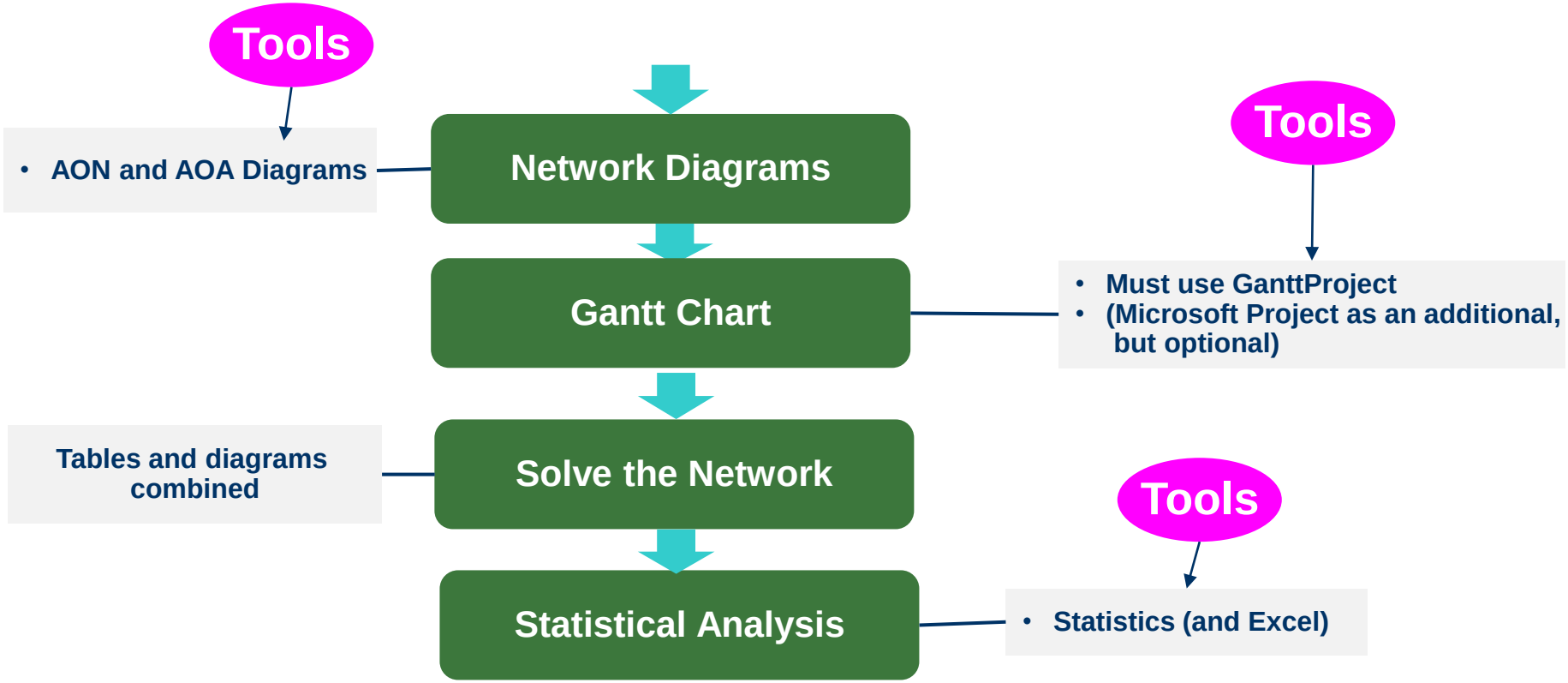
- Plan to manage change
- Communications plan
- Cost plan
- People plan
- Material planning
- WBS
- Deliveries vs. time
- Track and Reporting

WBS & RACI

Tools

- Hierachical tasks
- Subtasks
- Dependencies
- Sequence
- Who (& responsibilities)
- When

The Flow: Today's Topics



History of Methods

- **PERT**
  - Program Evaluation and Review Technique
  - Navy, Lockheed, etc 1958
  - R&D Projects
  - Statistical
- **CPM**
  - Critical Path Method
  - Dupont, etc
  - Construction
  - Deterministic

AON
(Activity on Node)

Terminology (1 of 2)

- **Activity** – tasks, resources, duration
- **Event** – result of one or more activities
- **Network** – combination of above
  - Drawn left-to-right
  - Connections represent predecessors
- **Path** - connected activities
- **Critical *Task*** -- if delayed, will delay the project
- **Critical *Path*** – at least one

Terminology (2 of 2)

- **Sequential Activities** - One activity must be completed before the next one can begin
- **Parallel Activities** - The activities can take place at the same time
- **Immediate Predecessor** - That activity that must be completed just before a particular activity can begin

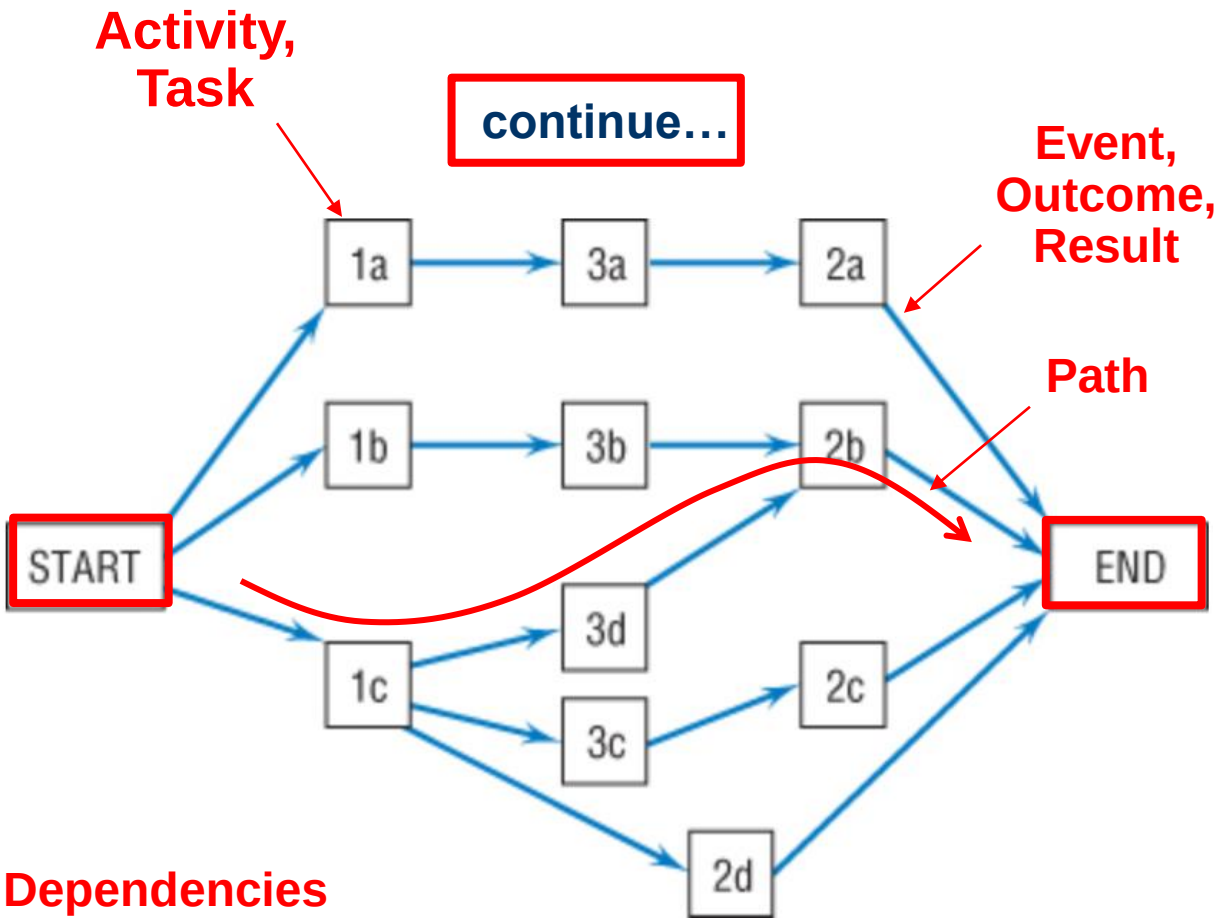
Introduction to Activity on Node (AON) Diagrams

Example: WBS

Tasks	Responsibility	Time (weeks)	Prec.	Resources
1. Contact Organizations				
a. Print forms	Secretary	6	–	Print shop
b. Contact organizations	Program manager	15	1.a	Word processing
c. Collect display information	Office manager	4	1.b	
d. Gather college particulars	Secretary	4	1.b	
e. Print programs	Secretary	6	1.d	Print shop
f. Print participants' certificates	Graduate assistant	8	–	Print shop
2. Banquet and Refre				
a. Select guest spea				
b. Organize food				er
c. Organize liquor				of Liquor Control
d. Organize refreshr				asing
3. Publicity and Promotion				
a. Send invitations	Graduate assistant	2	–	Word processing
b. Organize gift certificates	Graduate assistant	5.5	–	
c. Arrange banner	Graduate assistant	5	1.d	Print shop
d. Contact faculty	Program manager	1.5	1.d	Word processing
e. Advertise in college paper	Secretary	5	1.d	Newspaper
f. Class announcements	Graduate assistant	1	3.d	Registrar's office
g. Organize posters	Secretary	4.5	1.d	Print shop
4. Facilities				
a. Arrange facility for event	Program manager	2.5	1.c	
b. Transport materials	Office manager	.5	4.a	Movers

From WBS…

… to AON (PERT)



Good to show Task Dependencies

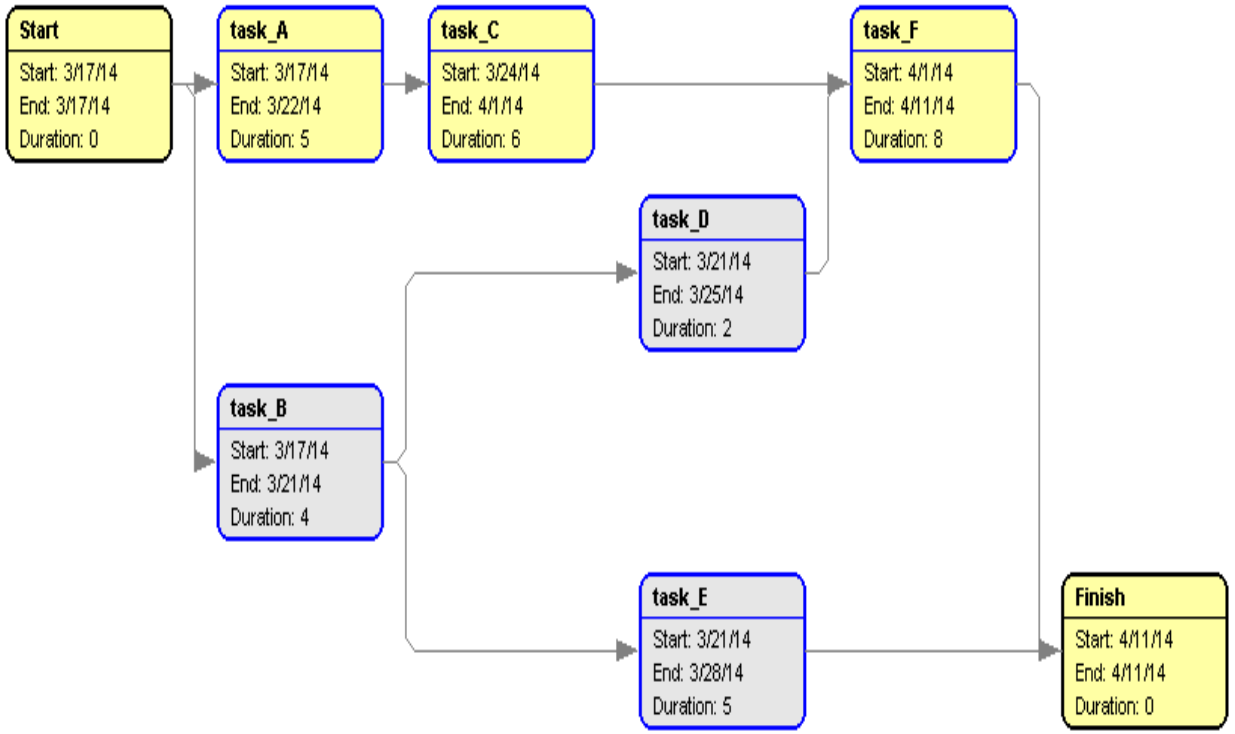
Constructing the AON

- Begin with START activity
- Add activities without precedence as nodes
  - There will always be one
  - May be more than one
- Add activities that have those activities as precedence(s)
- Continue

Example: Create an AON Chart from WBS

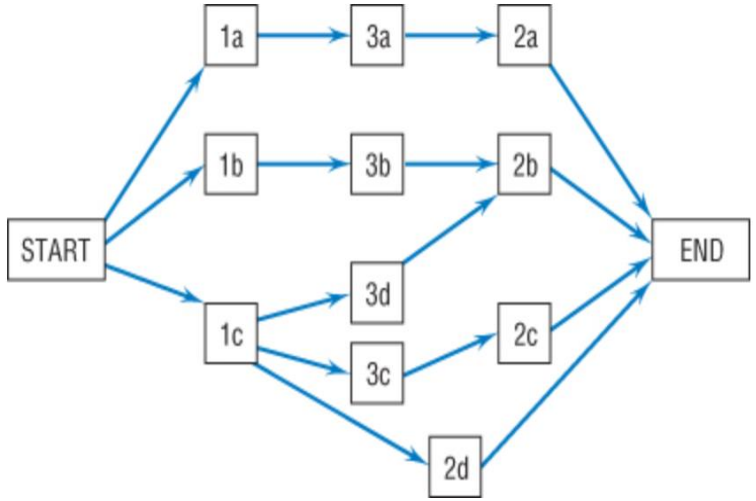
aka PERT chart

Tasks	*Precedence*	*Time*
a	—	5 days
b	—	4 days
c	a	6 days
d	b	2 days
e	b	5 days
f	c,d	8 days



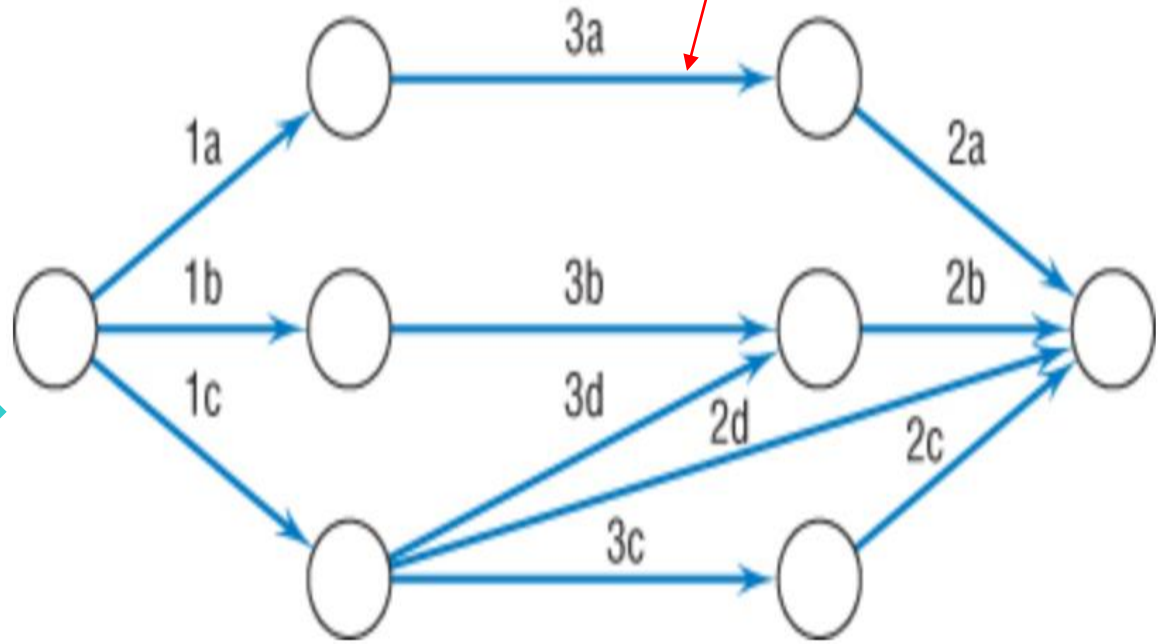
Introduction of Activity on Arrow (AOA)

activity



From AON (or WBS)…

… to AOA

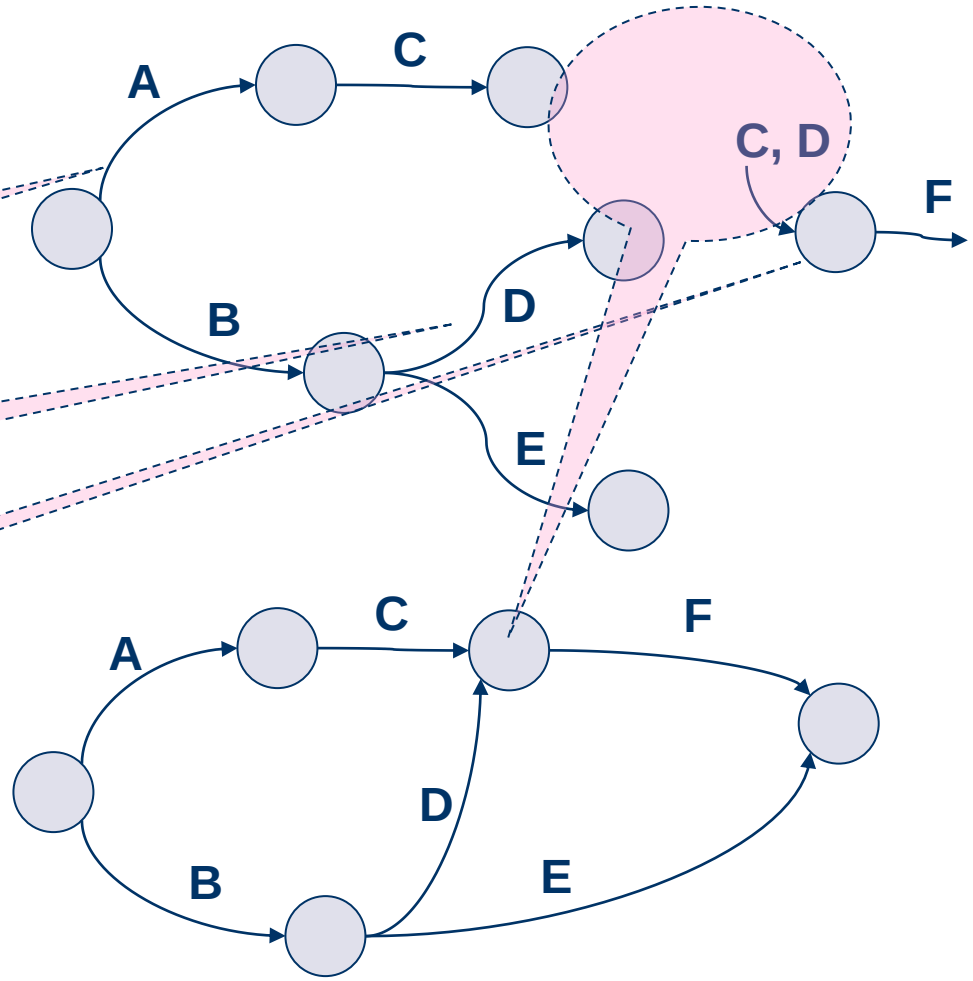


Example: From WBS to an AOA Chart

Tasks	Precedence	Time
a	—	5 days
b	—	4 days
c	a	6 days
d	b	2 days
e	b	5 days
f	c,d	8 days

fork

join



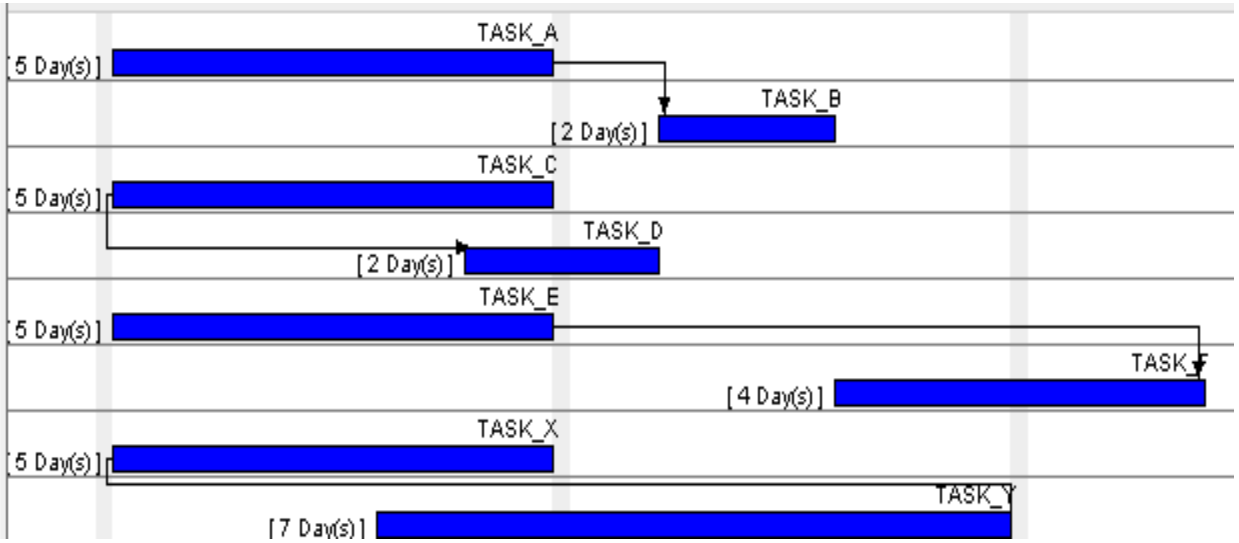
Precedence Diagramming

See “Example Precedence.gan” in Module #8

- Finish to start (FS) – example: frame first ***then*** pour concrete
- Start to start (SS) – e.g. start after some progress of known duration.
- Finish to finish (FF) – e.g. be ready but still work to be done
- Start to finish (SF) – ignore!

Can be parallel

TASK_A	3/10/14	3/14/14	5	
TASK_B	3/18/14	3/19/14	2	FS
TASK_C	3/10/14	3/14/14	5	
TASK_D	3/14/14	3/17/14	2	SS
TASK_E	3/10/14	3/14/14	5	
TASK_F	3/20/14	3/25/14	4	FF
TASK_X	3/10/14	3/14/14	5	
TASK_Y	3/13/14	3/21/14	7	SF



Examples: Start-to-Start

- Task 1: Assemble scaffolding
- Task 2: Paint exterior

Task 1 needs to start before Task 2 starts, but both can run in parallel

https://www.tacticalprojectmanager.com/start-to-start-dependency/

Examples: Finish-to-Finish

- **First task**: Paint the walls
- **Second task**: Do the floors

Both need to be done to be able to move-in. Can be done in parallel, with the only dependency to finish together.

- **First task**: Connect to the Internet
- **Second task**: Enable a remote monitor in extension mode

Gantt: Advantages and Disadvantage

Advantages

Easy-to-read method to know the current status
Shows planned and actual progress
Provide a picture of the current state of a project

Disadvantages

Difficult to follow complex projects

Network Scheduling Advantage

- Consistent framework
- Shows
  - Interdependences
  - when resources are needed
  - which task can be run parallel
- Ensures proper communication
- Determines
  - expected completion date
  - critical activities
  - activities which can be delayed
  - start dates
- Shows which task must be coordinated
- Allows probabilistic estimates

Your *boss* likes Gantt Charts

Most useful *to you* are the PERT/CPMs

Review

Q&A 8a

1. Of the choices below, what has the most significant *influence* on the management of projects?

a. GANTT chart
b. The Schedule
c. PERT chart
d. Organization chart

2.The series of connected activities or events between any two events in a network is called the _____.

a. path
b. event
c. network
d. Stream

3.What is more common in business project management today?

a. Activity on Arc (AOA)
b. Activity on Path (AOP)
c. Activity on Arrow (AOA)
d. Activity on Node (AON)

4. The GANTT Chart shows which of the following?

a. Activity on Arc (AOA)
b. Activity on Arrow (AOA)
c. Activities on Bars
d. Activity on Node (AON)

5. A node or event with duration of 0 days is a(n) _____.

a. milestone
b. error
c. short term activity (less than 1 day)
d. zero sum game

Next Topic

Learning Rate Recap

Mid-Term Exam Stats & Solutions

7:15

Entrepreneurship & Innovation

Network Diagrams & Gantt Charts

8:15

Network Solutions & Critical Path

7:50pm (25min)

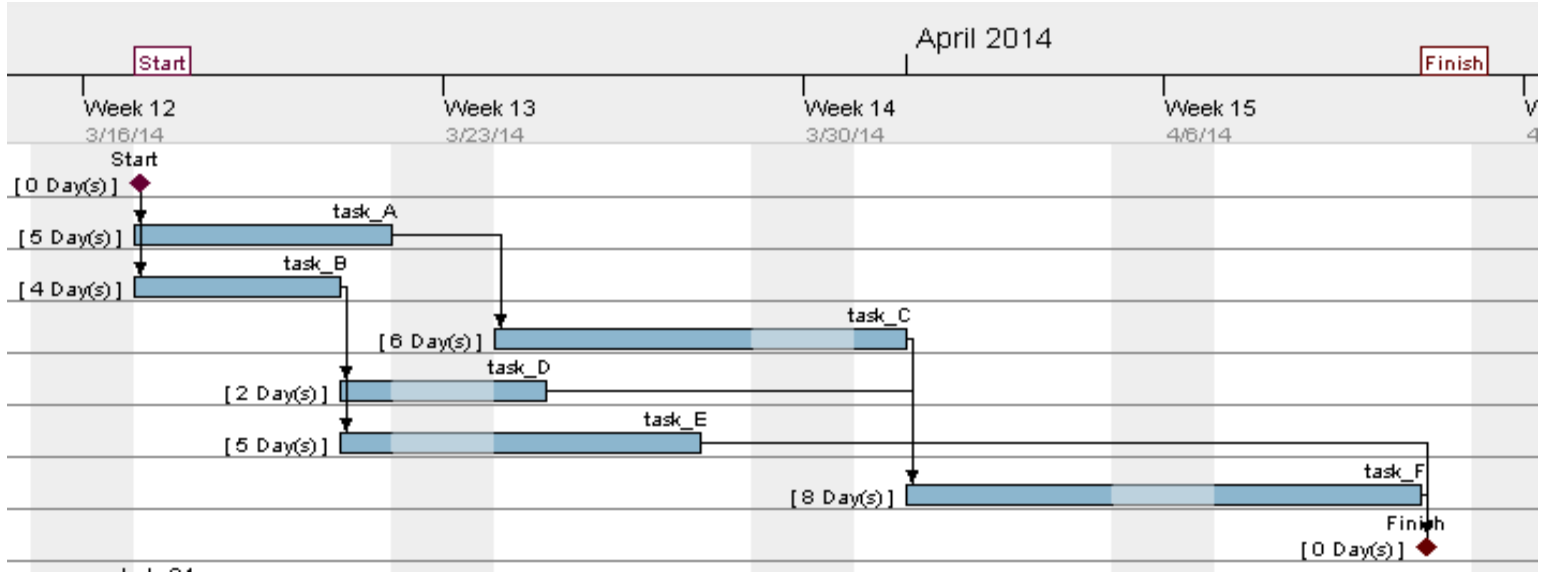
Project Related & Examples

8:20pm (70min)

Gantt Chart

Tasks	Precedence	Time
a	—	5 days
b	—	4 days
c	a	6 days
d	b	2 days
e	b	5 days
f	c,d	8 days

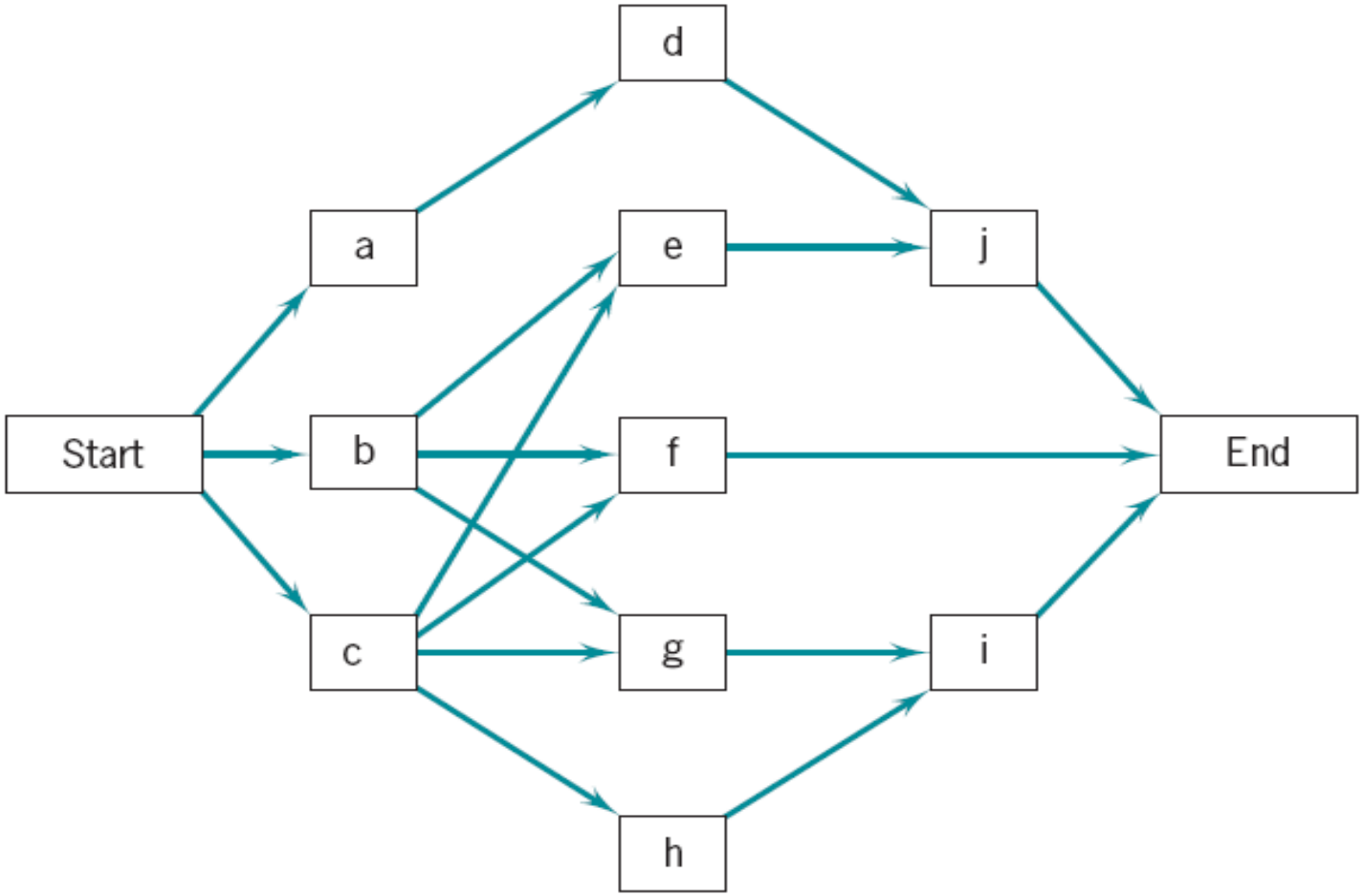
ID	Name	Begin date	End date	Dura...		Predecessors
0	Start	3/17/14	3/17/14	0		
1	task_A	3/17/14	3/21/14	5		0
3	task_B	3/17/14	3/20/14	4		0
4	task_C	3/24/14	3/31/14	6		1
5	task_D	3/21/14	3/24/14	2		3
6	task_E	3/21/14	3/27/14	5		3
7	task_F	4/1/14	4/10/14	8		4,5
8	Finish	4/11/14	4/11/14	0		6,7



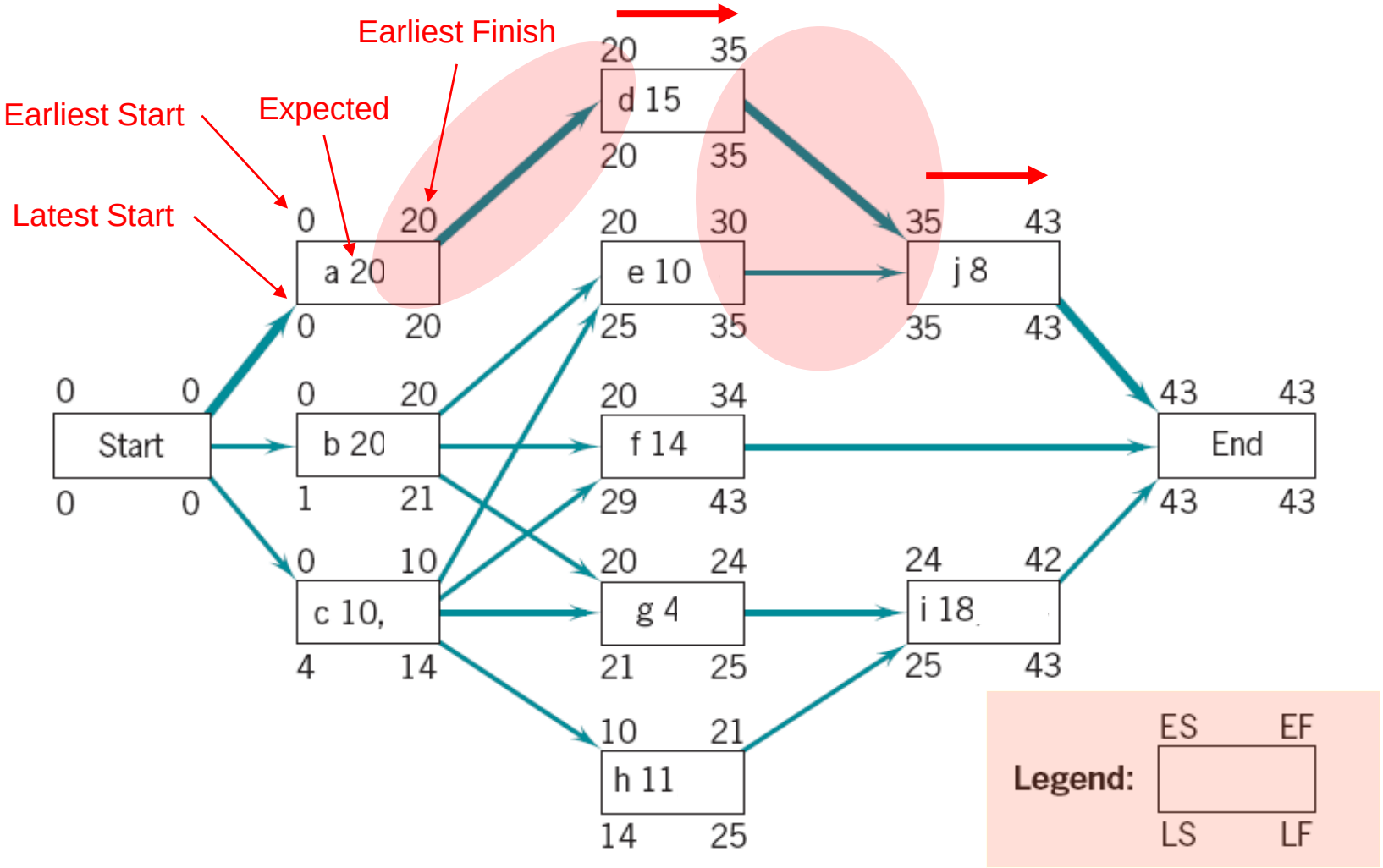
Solving the Network
Step 1: start with WBS

Activity	*Time*	*Immediate Predecessor Activities*
a	20	—
b	20	—
c	10	—
d	15	a
e	10	b, c
f	14	b, c
g	4	b, c
h	11	c
i	18	g, h
j	8	d, e

Step 2: Draw the AON Network

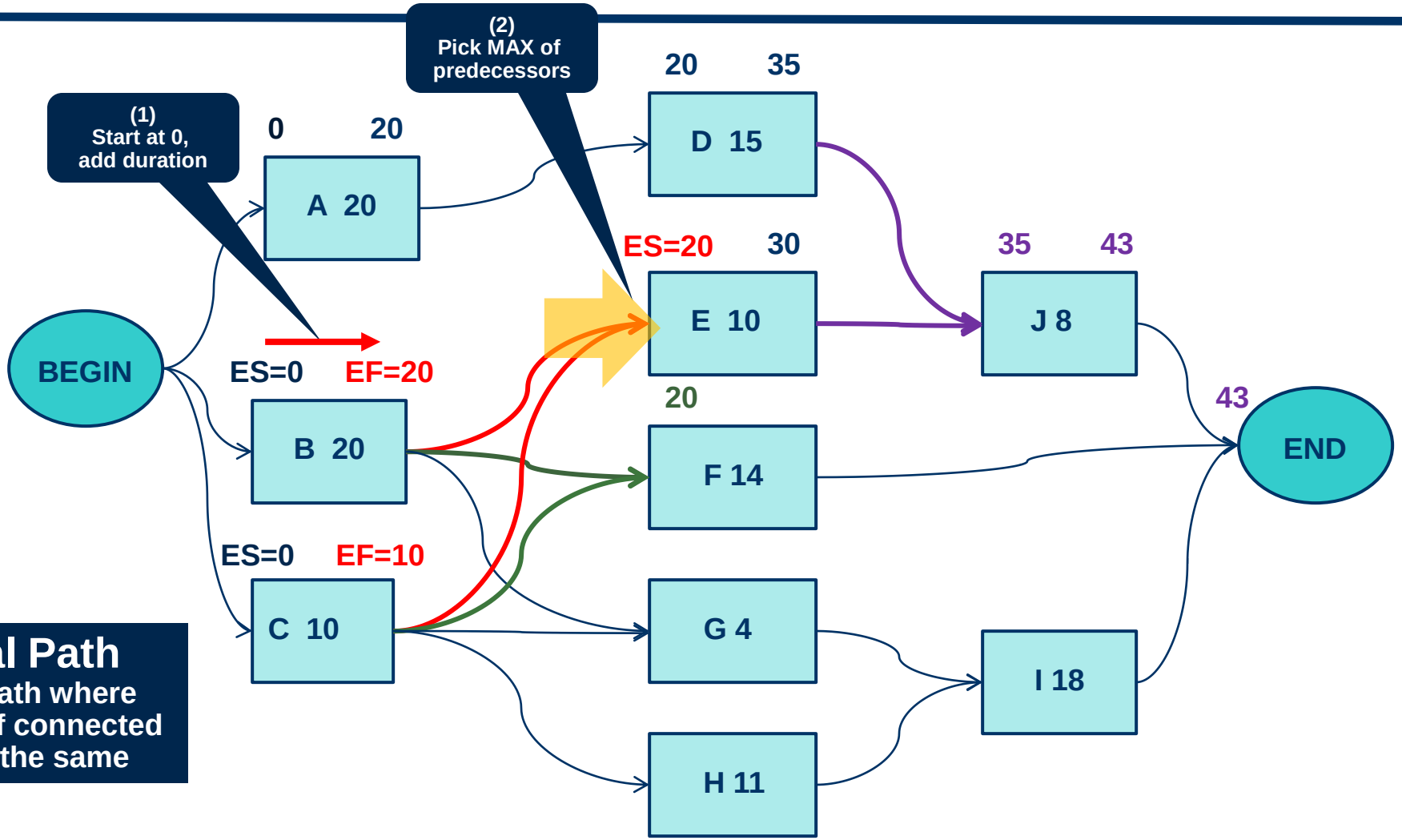


Step 3: Add Task Durations



Step 4: Solve "forwards"

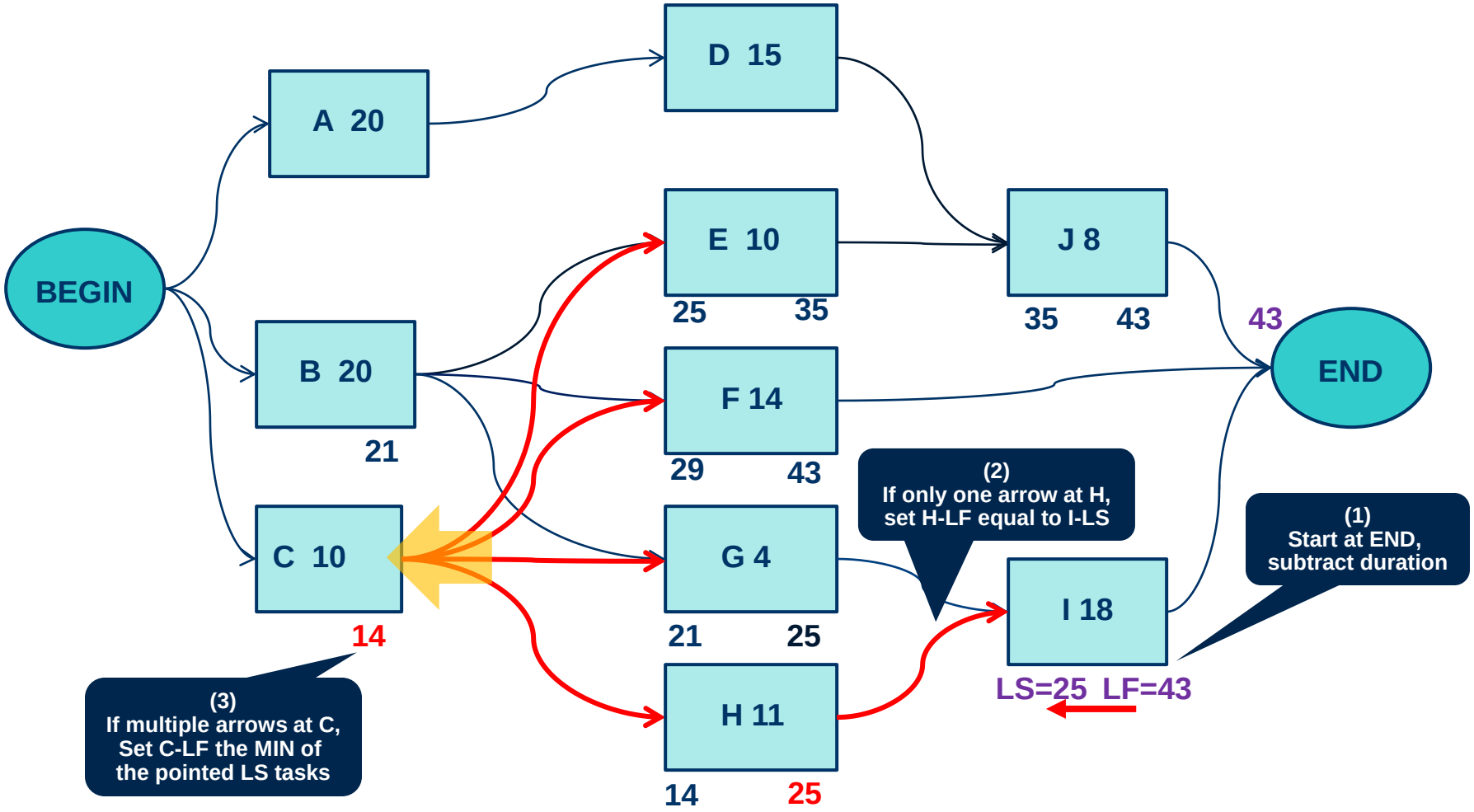
Forward RULE
Activity X cannot start until all paths LEADING TO it have been completed



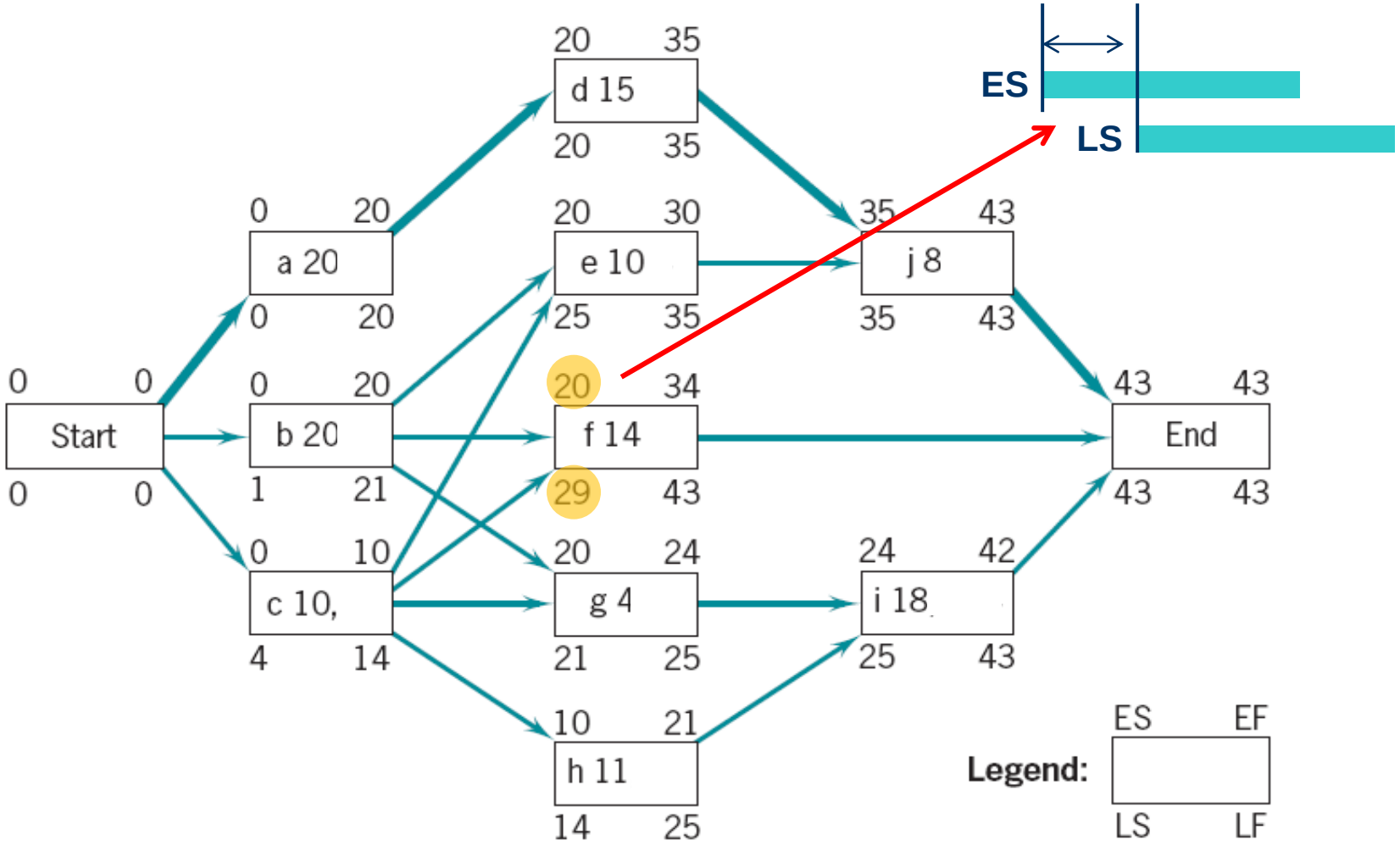
Critical Path
Longest path where EF and ES of connected tasks are the same

Step 5: Solve "backwards"

Backwards RULE
Activity X cannot finish (LF)
later than the pointed late starts (LS)



Definition of Slack (Float) Time

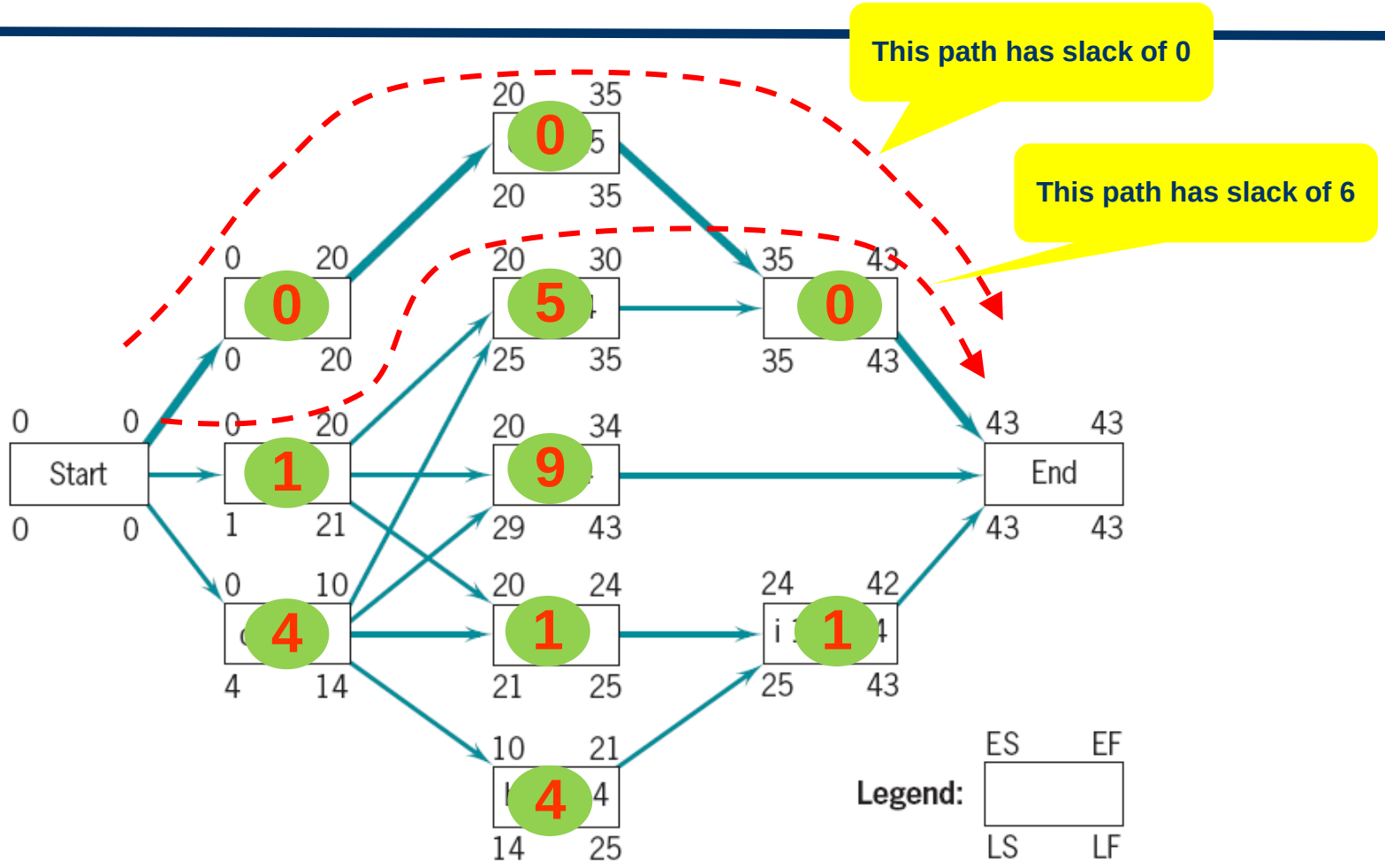


Step 8: Fill-in the Slack Table

Activity	*LS*	*ES*	*Slack*
a	0	0	0
b	1	0	1
c	4	0	4
d	20	20	0
e	25	20	5
f	29	20	9
g	21	20	1
h	14	10	4
i	25	24	1
j	35	35	0

Table 8-3

Critical Path (can be more than one!)



Parenthesis: time estimates

- It is vital to be meticulously *honest* in estimating time required per task
- Do not *sand-bag*!
- **It is not vital to be *accurate*.** *How much is pi (π)?*
- The *sum-of-parts* is typically GREATER than the total: so, step back and rethink.

Review

Q&A 8b

1. It is vital to good project management to be meticulously _____ in estimating the time required to complete each of the various tasks included in the project.

a. Quick
b. honest
c. technical
d. Managerial

2. _____ on the Critical Path _____.

a. No activities; have zero slack
b. The majority of activities; have some slack
c. All activities; have zero slack
d. The majority of activities; are full of slack

Next Topic

Learning Rate Recap

6:00pm (10min)

Mid-Term Exam Stats & Solutions

6:10pm (20min)

7:15

Entrepreneurship & Innovation

Network Diagrams & Gantt Charts

Network Solutions & Critical Path

8:15

Project Related & Examples

8:25pm (65 min)

Project Report

1. Expectations of Draft #3 → March 12th (after the break)
2. Risk Analysis (Review)
3. Network Diagrams (New Material)
4. Discussions

Next Steps for your Project: Towards Draft #3

1. *Consider my feedback* ***– significantly improve READABILITY and FORMAT!***
2. *Add missing sections*
3. *Add Risk Assessment* (Part of your Financial Plan)
   - Identify risks, use SWOT
   - Create a table with RPN (risk priority number)
   - Map to a 3x3 or 5x5 matrix
   - Assess risk and comment of your back-up plans
4. Add sections "Critical Success Factors" and "Assumptions" based on (3)
   - Just your high-level summary!
5. Create first draft of PERT Chart (based on next Class)
6. Create first draft of Gantt Chart (based on next Class) – must use GanttProject.exe

From Class #7 (should have been added by now)

Required for Draft #3

NOTE: Using the GanttProject is required for the course and your Term Project.

See links in Study Guide

“Good” examples: SWOT, FMEA

From Class #7

Strengths	Weaknesses
• Engineers with deep technical expertise • Easy to pivot with small nimble team • System does not rely on cloud connectivity	(W1) No access to cloud computing power (W2) Cannot rapid prototype mechanical housing ~~(W3) No internal exercise science expertise~~
Opportunities	**Threats**
• Using off-the-shelf hardware components • Friends with trainers/gym owners • Use of iPhone neural engine accelerates algorithm development time	(T1) Contract manufacturer is unreliable in quality or schedule (T2) Pandemic shutdowns prevent working in a gym (T3) The physics of exercise movements are not well understood enough to create useful algorithm

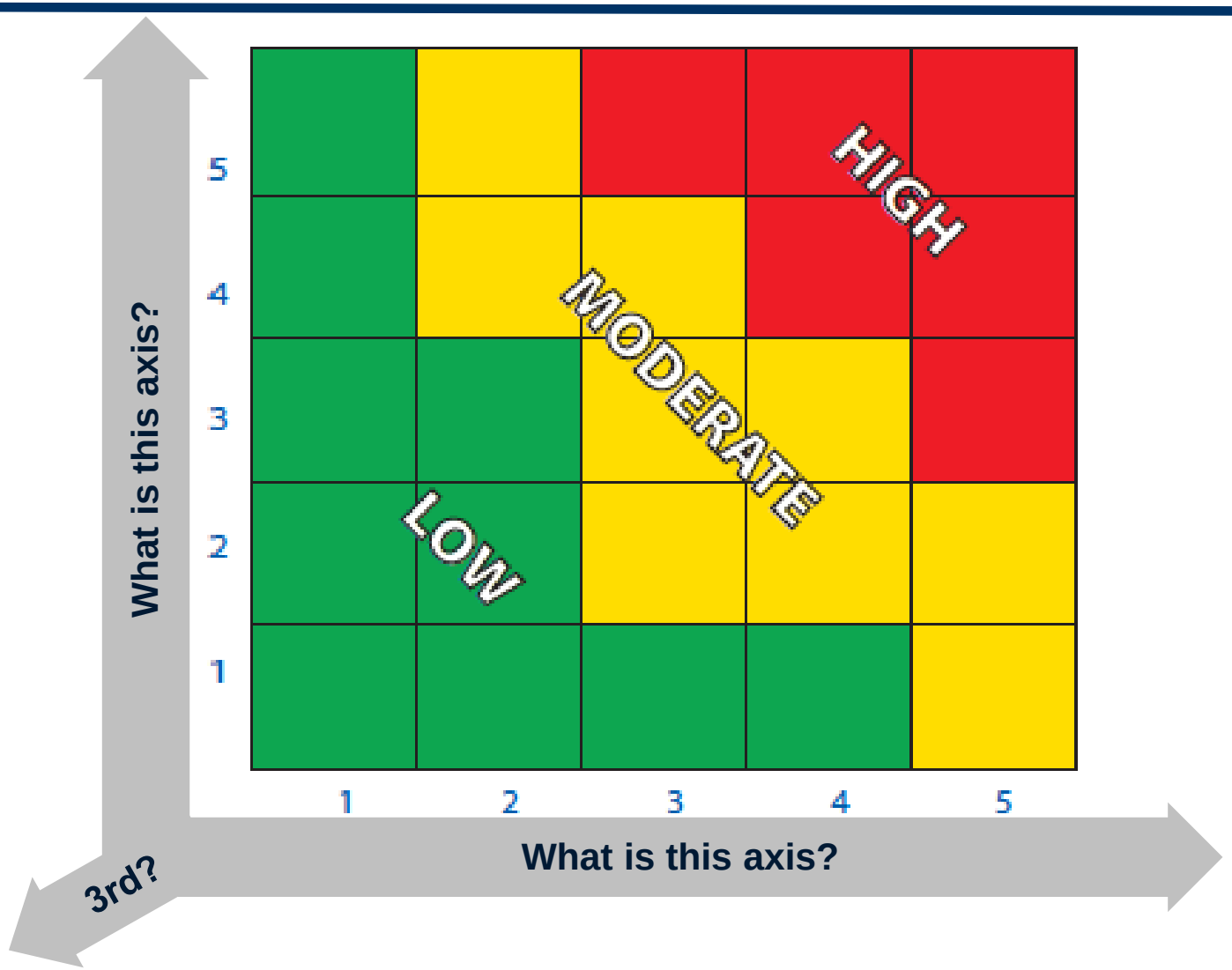
Threat	Severity	Likelihood	Inability to Detect	RPN
(W1) Not enough computing power unless using cloud computing	6	3	2	36
(W2) Delays in iterating on mechanical housing	2	9	2	36
(T1) CM is unreliable in quality/schedule	3	4	7	84
(T2) Shutdowns prevent working in gym	4	7	8	224
(T3) Not possible to create useful algorithm	10	5	2	100

Additions?
Improvements?

Risk Matrix (5x5)

Risk Priority Number
(RPN)

S?
L?
D?

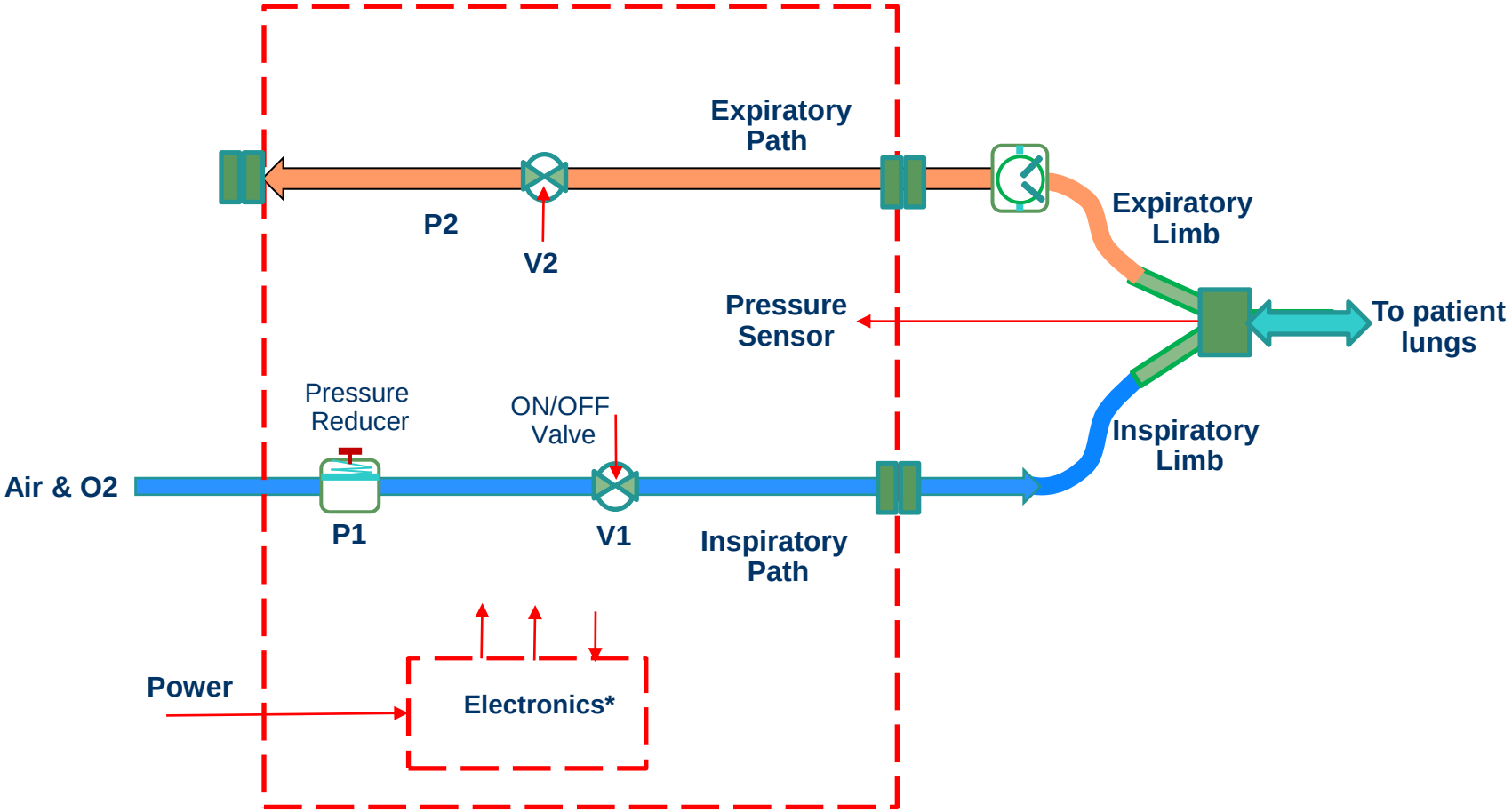


The Ventilator Example (Model for your Project)

1. **Requirements**
2. **Technical Approach**
   - Mechanical
   - Timing Diagram
   - State Diagram
3. **WBS Table**
4. **Risk Assessment**
5. **AON Network (PERT Chart)**
6. **Gantt Chart**
7. **Solving the Network**
   - Forwards
   - Backwards
8. **Slacks**
9. **Critical Path**
10. **Probabilities of completion**

Step #1: Technical Approach

Review



Tasks and WBS

Review

ID	Task Name	Duration	Predecessors
	Mechanical		
ID1	Select Valves, Hoses	3	
ID2	Design, fittings and assemble	8	ID1
	Electronics		
ID4	MCU and Interface to valves	3	ID1
ID5	Create 1st prototype	10	ID4
ID7	Create firmware	9	ID1; ID5
ID9	Integrate on the bench	4	ID2; ID5
ID6	*PROTO READY*	*MILESTONE*	ID9
ID8	Create GUI	12	ID6
	Test and Qualify		
ID11	First Hospital Test	2	ID7
ID12	*TRIAL READY*	*MILESTONE*	ID8; ID11
ID13	Live subject testing	2	ID12

See Ventilator example in Module#8

(included in the "Shell")

Step #4a: Risk Assessment

Review

Strengths (internal)

S1. Experience in electromechanical designs and embedded code development
S2. Extensive reusable code and tools
S3. Multidiscipline skills including bioengineering

Weaknesses (internal)

W1. Limited mobility for integration (virus)
W2. Timeline is very tight
W3. Company has not a committed budget
W4. Doctor team member not easily accessible

Opportunities (external)

O1. Using hospital air is simpler than making air pumps.
O2. Use off-the-shelf components (valves, regulators, sensors)
O3. Access to hospital labs and personnel

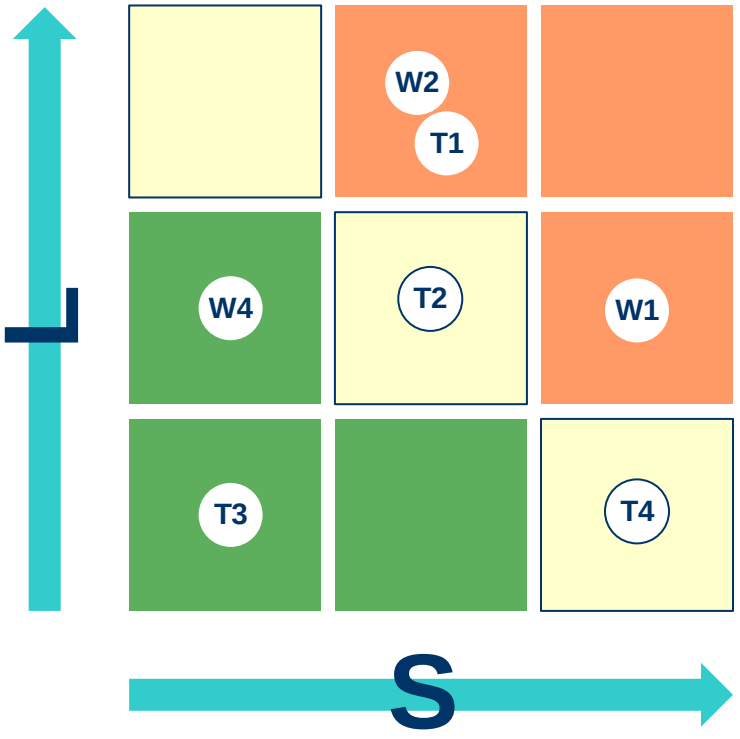
Threats (external)

T1. Delays in deliveries of components
T2. Non-committed budget
T3. Quality of purchased components varies
T4. Fresh Pond campus security stops access to the building

Review

Step #4b: RPN and Risk Matrix

Weaknesses & Threats	S	L	D	RPN
W1. Limited mobility for integration (virus)	3	2	3	**18**
W2. Timeline is very tight	2	3	2	**12**
W4. Doctor team member not easily accessible	1	2	1	**2**
T1. Delays in deliveries of components	2	3	1	**6**
T2. Non-committed budget	2	2	2	**8**
T3. Quality of purchased components varies	1	1	1	**1**
T4. Fresh Pond campus security stops access	3	1	2	**6**



Mediation Plan

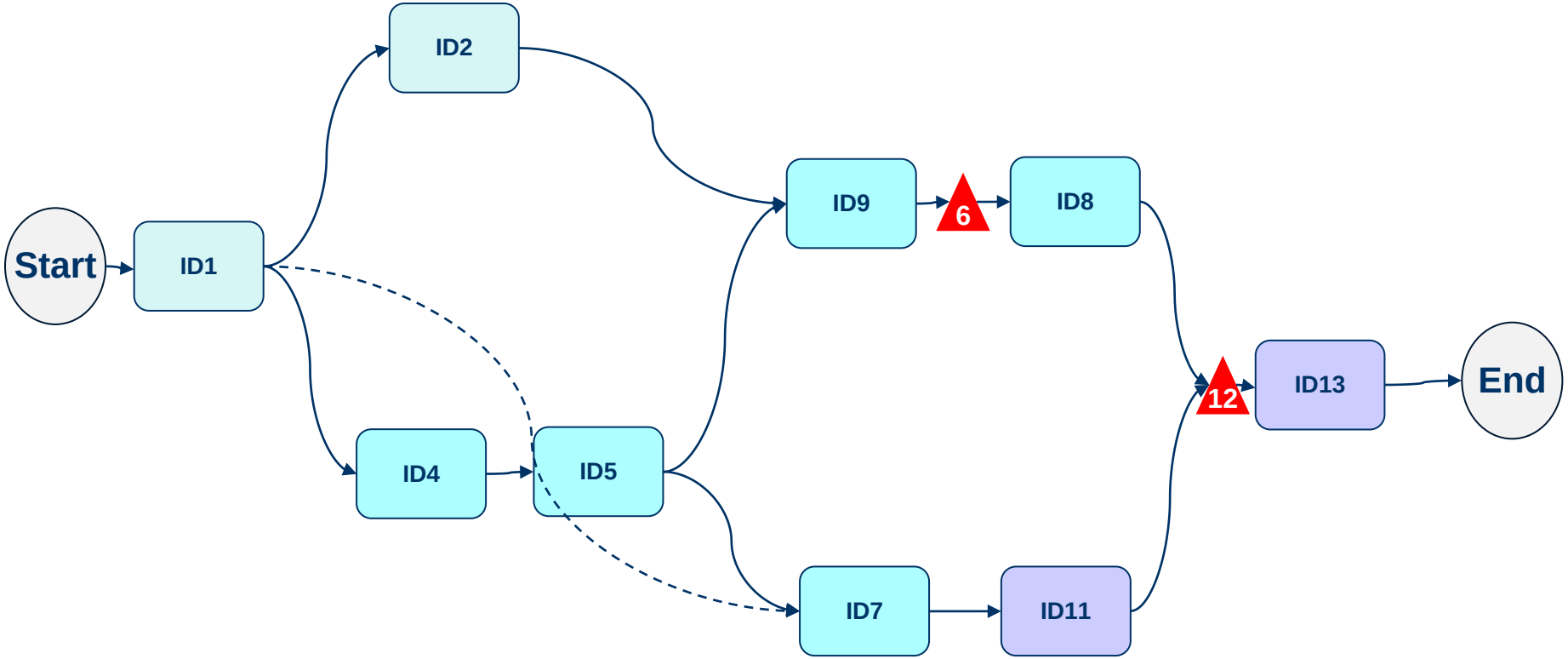
- W1: work remotely & Zoom twice a day
- W2: reduce features to bare minimum, acquire COTS* components
- T1: find second source(s)

*COTS stands for: Commercial Off-The Shelf

New Material

Step #5: Create AON from WBS

ID	Pre-decessors
ID1	
ID2	ID1
ID4	ID1
ID5	ID4
ID7	ID1; ID5
ID9	ID2; ID5
6	ID9
ID8	ID6
ID11	ID7
ID13	ID12
12	ID8; ID11



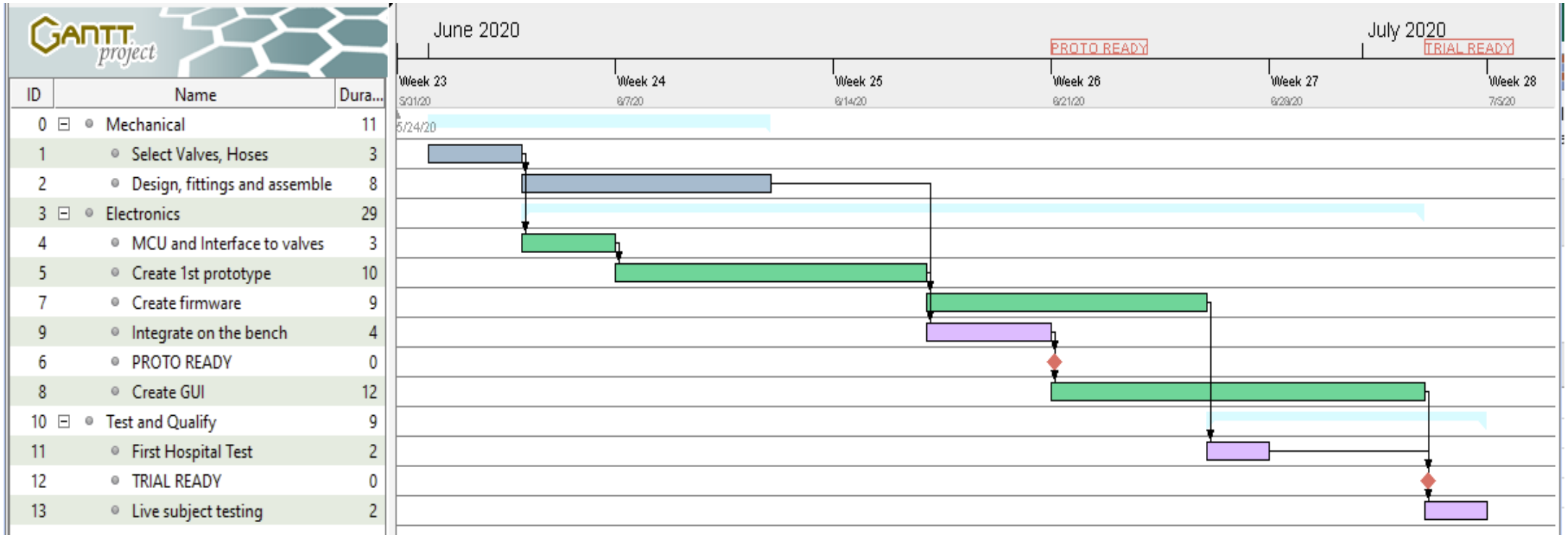
New Material

Step #7 : Create Gantt Chart

Enter all tasks, durations and predecessors to GranttProject

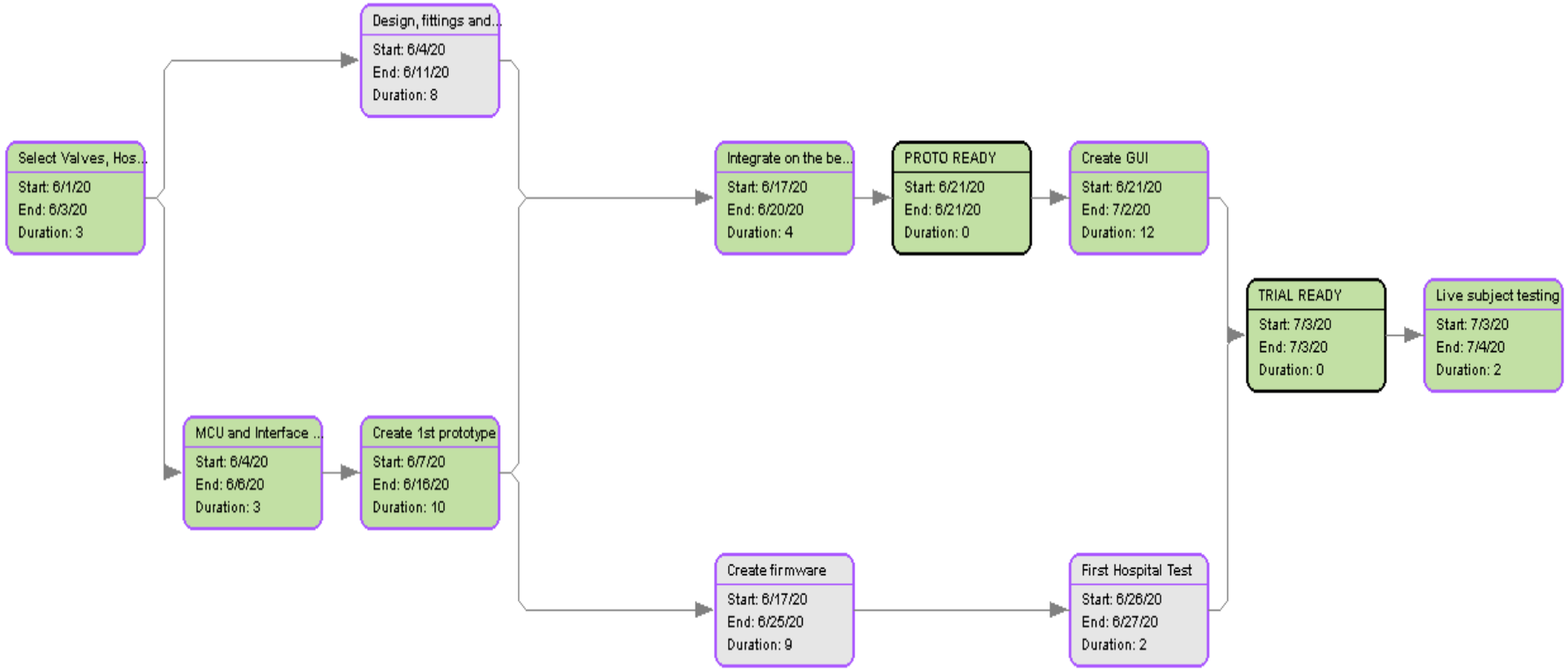
Vent Ex Gantt.gan

Vent Ex Gantt



New Material

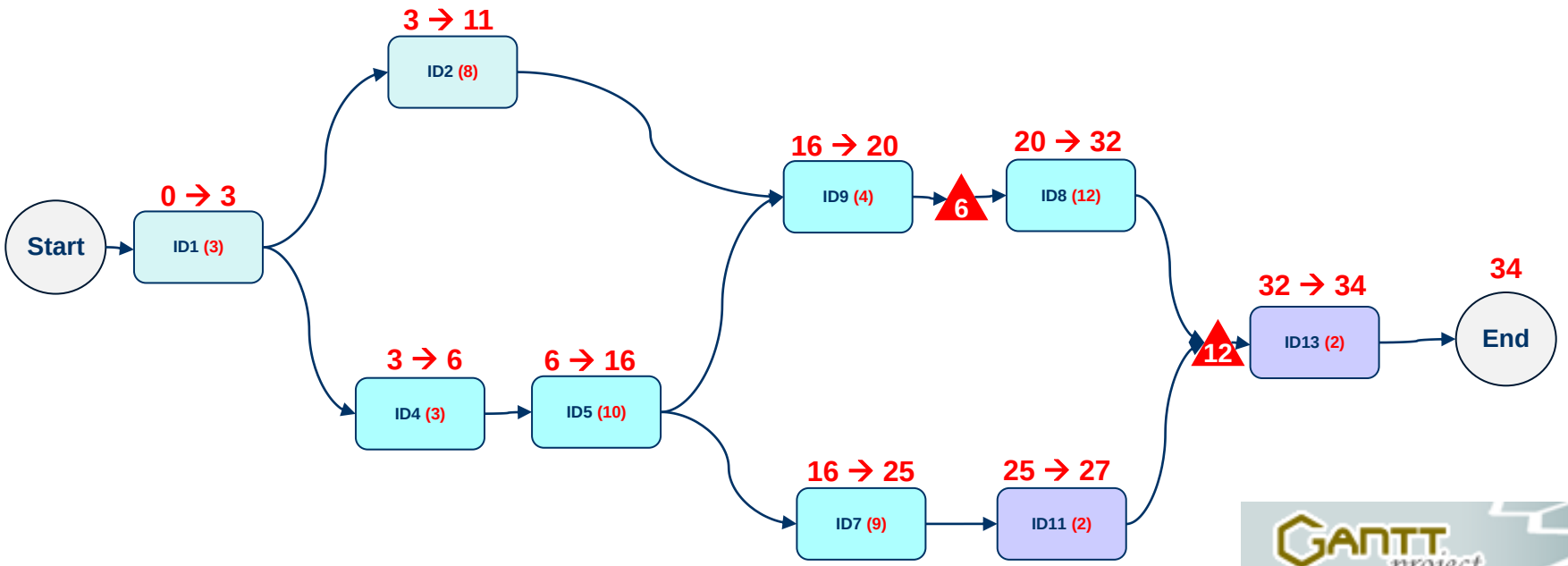
Step #8: Confirm Pert Cart



Cut and paste from GanttProject. Compare with Step #5

New Material

Step #9: Add Duration and Solve Forwards



Confirm End (34 days) with Gantt Chart

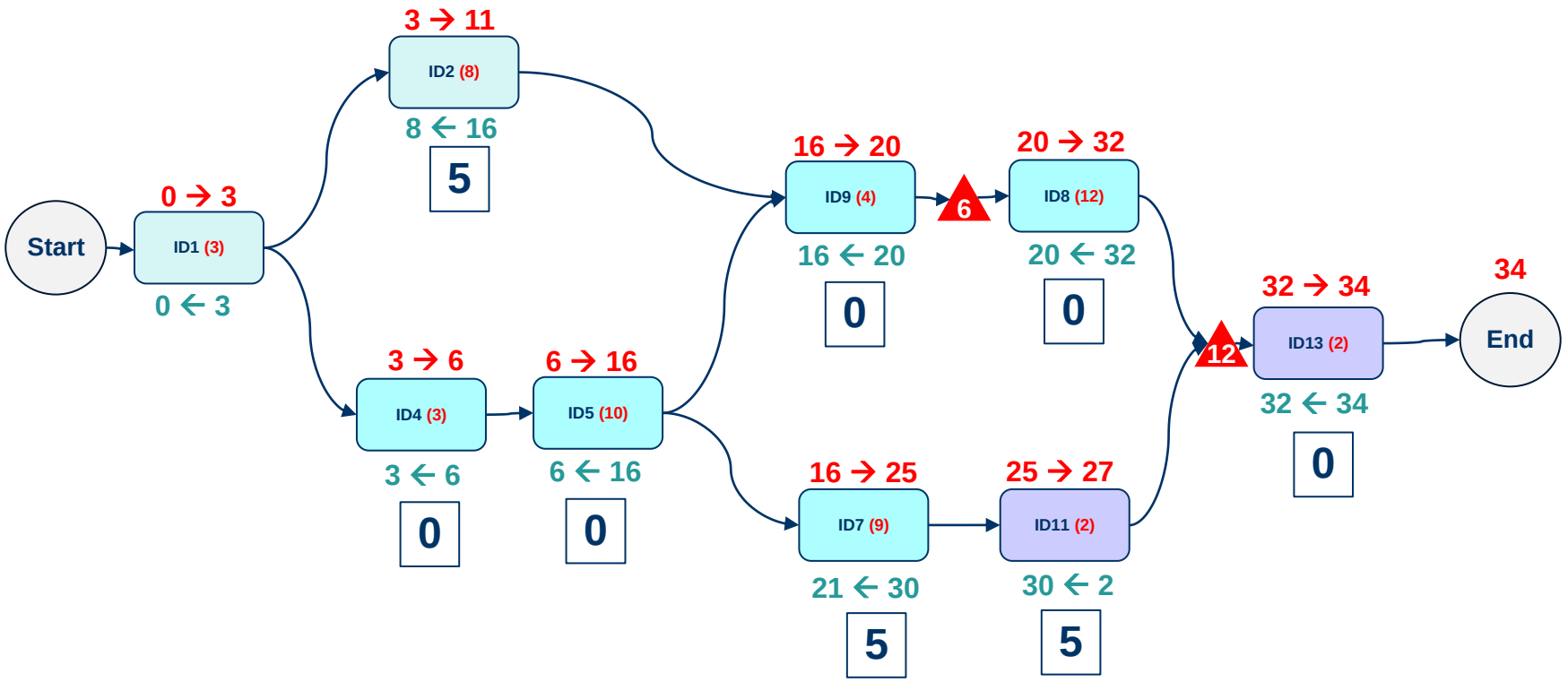
How did you do this?

GANTT project

ID	Name	Dura...
14	TOTAL	34
0	Mechanical	11
3	Electronics	29
10	Test and Qualify	9

New Material

Step #10: Solve backwards. Find Task Slacks



Which is the Critical Path?

Class Discussion on the Project

1. Make a VERY QUICK intro of your project, i.e. your SCOPE
2. Show your SWOT analysis and explain the Weaknesses and Threats
3. Show your 3x3 (or 5x5) Risk Matrix